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Model theoretical aspects of the logic of formal inconsistency **QLFI1**

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Abstract

With the aim of dealing with inconsistent databases, Carnielli, Marcos and de Amo introduced in 2000 a 3-valued *logic of formal inconsistency* (LFI) named **LFI1**, which turn out to be functionally equivalent to da Costa and D'Ottaviano's **J3**.

In this paper, we present a new approach to the semantics of triples, introduced by Coniglio and Silvestrini, in the context of the first-order version of **LFI1**, **QLFI1**. We define partial sets which serve as the main tool for constructing what we call “partial structures” which are essentially Tarskian structures except that predicate symbols are interpreted as partial sets. We endow partial sets with operations that allow us to algebraize them, from which we establish the notion of **QLFI1**-structure.

After proving soundness and completeness of the logic **QLFI1** with respect to **QLFI1**-structures, we present versions of several results from model theory adapted to the setting of **QLFI1**-structures.

1 Introduction

In his pioneering paper “A propositional calculus for inconsistent deductive systems” from 1948, Jaśkowski establishes the problem and criterion of paraconsistency (see [2, pg 140-141]). In the same paper, Jaśkowski proposes the so called discursive (or discursive) logic as a possible solution to the problem posed by his teacher J. Łukasiewicz. In 1970, D'Ottaviano and da Costa proposed a 3-valued modal logic called **J3** as an alternative solution to Jaśkowski's problem (see [13]). They also presented a first-order version of this logic. Afterwards, in her Ph.D. thesis [9] and in a series of papers D'Ottaviano obtains several results concerning the model theory of first-order paraconsistent logic **J3** (see [11, 10, 12]). Together with Dunn's paper [14], this constitutes a pioneering work on 3-valued paraconsistent first-order model theory, which was subsequently rediscovered in the context of inconsistent databases and the theory of quasi-truth (see [2, Sect. 7.9 of Chap. 7 and Sect. 9.3 of Chap. 9]). The basic language of **J3** is formed by a disjunction, a paraconsistent negation and a “possibility” operator ∇ .

The logic **J3** has independently been reintroduced by several authors with different signatures and different motivations. With the aim of dealing with inconsistent databases, Carnielli, Marcos and de Amo introduced in 2000 a

3-valued *logic of formal inconsistency* (LFI) named **LFI1** which they proved to be functionally equivalent to **J3** (see [3]). That is, the primitive operators of both matrices are interdefinable, where the signature of **LFI1** includes an inconsistency $\bullet$ operator instead of ∇ . In turn, in 2004, Batens and De Clercq introduced, by means of a Hilbert calculus, a first-order paraconsistent logic called **CLuNs** (see [1]). The propositional fragment of **CLuNs** was shown to be sound and complete with respect to the matrix of **J3**.

On the other hand, the notion of quasi-truth introduced in 1986 by Mikenberg et al. (see [16]) was then generalized by Coniglio and Silvestrini in [19, 6]. This generalization is axiomatized by a first-order paraconsistent logic called **LPT1**, whose propositional fragment is characterized by a 3-valued matrix logic called **MPT**. As proven in [6], the logic **MPT** coincides (up to language) with **J3**, thus constituting a new variant of **J3** which again arises independently in a different context and from a different motivation. Consequently, **LPT1** coincides (up to language) with the first-order version of **J3** studied by D'Ottaviano and also with the semantics for first-order **LFI1** proposed in [7] for inconsistent databases. However, the semantics of triples proposed in [6] for **LPT1** is of a different nature than those of the other two approaches.

In this paper, we present a new approach to the semantics of triples in the context of the first-order version of **LFI1**, **QLFI1**. We define partial sets which serve as the main tool for constructing what we call “partial structures” which are essentially Tarskian structures except that predicate symbols are interpreted as partial sets. We endow partial sets with operations that allow us to algebraize them, from which we establish the notion of **QLFI1**-structure. After proving soundness and completeness of the logic **QLFI1** with respect to **QLFI1**-structures, we present versions of several results from model theory adapted to the setting of **QLFI1**-structures. Here we follow a similar path to the one followed in [5].

2 Propositional Logic LFI1

The propositional logic **LFI1** was introduced in [3] by Carnielli, Marcos, and de Amo, with the aim of modeling processes in evolutionary databases. As we said, this same logic had previously been introduced, under different presentations, by other authors.

In [19] (see also [6]), Coniglio and Silvestrini introduce the logic **LPT**, seeking to establish a three-valued matrix associated with pragmatic structures, which serve as a pathway to a generalization of the theory of *quasi-truth* previously developed by Mikenberg et al. (see [16]). It has also been shown that **LPT** and **J3** (and therefore **LFI1**) are functionally equivalent.

Initially, the signature of **LFI1** included the primitive symbol $\bullet$, intended to indicate inconsistencies in the database via the equivalence

$$\bullet A \leftrightarrow (A \wedge \neg A).$$

In [17], the matrix with the white ball and the axiomatization we adopted (see also the book [2], Def. 3 and Th. 3) were presented. Instead of using $\bullet$ as a primitive, the operator $\circ$, called the *consistency* operator, was employed; both operators are interdefinable, since $\circ x = \neg \bullet x$. This same consistency operator

was also defined in [15, p. 269], where we can already find, so to speak, the embryo of **LFII**.

In the recent book [2] by Carnielli and Coniglio, a new approach to **LFII** is presented, taking the symbol $\circ$ as primitive and introducing a new Hilbert-style axiomatization of the associated calculus. This most recent development of **LFII** is the one we adopt in what follows. Let us consider the propositional language $\mathcal{L}_\Theta$ with signature $\Theta = \{\wedge, \vee, \rightarrow, \neg, \circ\}$.

Definition 2.1 Let $\mathcal{M}_{\mathbf{LFII}} = \langle M, D \rangle$ be the matrix over the signature Θ with domain $M = \{1, 1/2, 0\}$ and designated values $D = \{1, 1/2\}$, where the operators are defined by the following truth tables:

$\wedge$	1	$1/2$	0
1	1	$1/2$	0
$1/2$	$1/2$	$1/2$	0
0	0	0	0

$\vee$	1	$1/2$	0
1	1	1	1
$1/2$	1	$1/2$	$1/2$
0	1	$1/2$	0

$\rightarrow$	1	$1/2$	0
1	1	$1/2$	0
$1/2$	1	$1/2$	0
0	1	1	1

	$\neg$	$\circ$
1	0	1
$1/2$	$1/2$	0
0	1	1

The logic $\mathbf{LFII} = \langle \mathcal{L}_\Theta, \models_{\mathbf{LFII}} \rangle$ is the logic associated with the matrix $\mathcal{M}_{\mathbf{LFII}}$, i.e., for every $\Gamma \cup \{A\}$, we define

$$\Gamma \models_{\mathbf{LFII}} A$$

if and only if, for every $\mathcal{M}_{\mathbf{LFII}}$ -valuation v we have $h[\Gamma] \subseteq \{1, 1/2\}$ only if $h(A) \in \{1, 1/2\}$. If $\Gamma = \emptyset$, we write $\models_{\mathbf{LFII}} A$ and say that A is an **LFII**-tautology.

Remark 2.2 The connectives $\wedge$, $\neg$, and $\circ$ can be taken as primitive. As we shall see, $\vee$ and $\rightarrow$ can be seen as defined connectives.

We introduce the following connectives as abbreviations:

$$\begin{aligned} \sim A &:= \neg A \wedge \circ A \\ A \vee B &:= \neg(\neg A \wedge \neg B) \\ A \rightarrow B &:= (\neg A \wedge \circ A) \vee B \\ A \leftrightarrow B &:= (A \rightarrow B) \wedge (B \rightarrow A) \\ A \equiv B &:= (A \leftrightarrow B) \wedge (\circ A \leftrightarrow \circ B) \end{aligned}$$

with their corresponding truth tables given as follows:

	$\sim$
1	0
$1/2$	0
0	1

$\leftrightarrow$	1	$1/2$	0
1	1	$1/2$	0
$1/2$	$1/2$	$1/2$	0
0	0	0	1

$\equiv$	1	$1/2$	0
1	1	0	0
$1/2$	0	$1/2$	0
0	0	0	1

Definition 2.3 Let **LFI1** be the Hilbert-style deductive system whose axiom schemata are:

$$(\mathbf{Ax1}) \quad \alpha \rightarrow (\beta \rightarrow \alpha)$$

$$(\mathbf{Ax2}) \quad (\alpha \rightarrow (\beta \rightarrow \gamma)) \rightarrow ((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \gamma))$$

$$(\mathbf{Ax3}) \quad \alpha \rightarrow (\beta \rightarrow (\alpha \wedge \beta))$$

$$(\mathbf{Ax4}) \quad (\alpha \wedge \beta) \rightarrow \alpha$$

$$(\mathbf{Ax5}) \quad (\alpha \wedge \beta) \rightarrow \beta$$

$$(\mathbf{Ax6}) \quad \alpha \rightarrow (\alpha \vee \beta)$$

$$(\mathbf{Ax7}) \quad \beta \rightarrow (\alpha \vee \beta)$$

$$(\mathbf{Ax8}) \quad (\alpha \rightarrow \gamma) \rightarrow ((\beta \rightarrow \gamma) \rightarrow ((\alpha \vee \beta) \rightarrow \gamma))$$

$$(\mathbf{Ax9}) \quad (\alpha \rightarrow \beta) \vee \alpha$$

$$(\mathbf{Ax10}) \quad \alpha \vee \neg \alpha$$

$$(\mathbf{Ax11}) \quad \circ \alpha \rightarrow (\alpha \rightarrow (\neg \alpha \rightarrow \beta))$$

$$(\mathbf{Ax12}) \quad \neg \circ \alpha \rightarrow (\alpha \wedge \neg \alpha)$$

$$(\mathbf{Ax13}) \quad \alpha \leftrightarrow \neg\neg\alpha$$

(Ax14) $\neg(\alpha \vee \beta) \leftrightarrow (\neg\alpha \wedge \neg\beta)$

$$(\mathbf{Ax15}) \quad \neg(\alpha \wedge \beta) \leftrightarrow (\neg\alpha \vee \neg\beta)$$

$$(\mathbf{Ax16}) \quad \neg(\alpha \rightarrow \beta) \leftrightarrow (\alpha \wedge \neg\beta)$$

and whose sole rule of inference is Modus Ponens (MP) $\frac{\alpha \quad \alpha \rightarrow \beta}{\beta}$.

We denote by $\vdash_{\mathbf{LFI1}}$ the consequence relation determined by the system **LFI1**.

Theorem 2.4 ([2, Th. 4.4.43]) For every $\Gamma \cup \{\psi\} \subseteq \mathcal{L}_\Theta$.

$$\Gamma \models_{\mathbf{LFI1}} \psi \quad \text{iff} \quad \Gamma \vdash_{\mathbf{LFI1}} \psi$$

Proposition 2.5 The following schemata are valid in **LFI1**.

$$\mathbf{T1} \quad \circ \neg \psi \equiv \circ \psi$$

$$\mathbf{T4} \quad \circ\psi \equiv \circ\psi \wedge (\psi \vee \neg\psi)$$

T2 $\neg\neg\psi \equiv \psi$

$$\mathbf{T5} \quad \circ\psi \wedge \neg \circ\psi \equiv (\psi \wedge \neg\psi) \wedge (\psi \wedge \circ\psi)$$

T3 $\circ\psi \equiv \circ\psi \wedge \circ\circ\psi$

T6 $\varphi \wedge \psi \wedge \circ(\varphi \wedge \psi) \equiv (\varphi \wedge \circ\varphi) \wedge (\psi \wedge \circ\psi)$

$$\mathbf{T7} \quad \psi \equiv \psi \wedge (\neg\psi \vee \circ\psi)$$

$$\mathbf{T8} \quad \neg(\varphi \wedge \psi) \wedge \circ(\varphi \wedge \psi) \equiv (\neg\varphi \wedge \circ\varphi) \vee (\neg\psi \wedge \circ\psi)$$

Proof. It is routine. ■

3 Partial Relations and partial structures

In order to define the semantical framework for **QLFI1**, the first-order version of **LFI1** to be introduced next, it will be convenient to consider partial relations, which were first introduced in the literature in **QCiore** (see [5]), defined in terms of triples. Note that we could have used the more standard structure of pairs of extension/anti-extension, as employed, for example, by Priest in the LP-models for the first-order version of the paraconsistent logic LP (see [18]). The triples approach was adopted here because it is closely related to the quasi-truth theory, as explained in the Introduction. This representation is quite clear from the conceptual point of view, keeping all the semantic information about the sentences (truth, falsity, and doubt) in separate components. Moreover, pragmatic structures provide a formal framework in which definitions and proofs become clearer and technically simpler to handle.

Denote by **3** the set $\{0, 1/2, 1\}$. Given a nonempty set U , a *partial set* or **3-set** over U is a function $A \in \mathbf{3}^U$. If A is a **3-set**, we denote it by

$$A = \langle A_{\oplus}, A_{\ominus}, A_{\odot} \rangle,$$

where $A_{\oplus} = A^{-1}(1)$, $A_{\ominus} = A^{-1}(0)$, and $A_{\odot} = A^{-1}(1/2)$.

Remark 3.1 Let A be a partial set in $\mathbf{3}^U$. Then the following hold:

- (1) $A_{\oplus} \cup A_{\ominus} \cup A_{\odot} = U$.
- (2) $A_{\oplus} \cap A_{\ominus} = \emptyset$, $A_{\oplus} \cap A_{\odot} = \emptyset$, and $A_{\ominus} \cap A_{\odot} = \emptyset$.

Conversely, if A, B , and C are subsets of a nonempty set U such that

- (3) $A \cup B \cup C = U$.
- (4) $A \cap B = \emptyset$, $A \cap C = \emptyset$, and $B \cap C = \emptyset$,

then there exists $X \in \mathbf{3}^U$ such that $X = \langle A, B, C \rangle$.

The idea of mapping a domain into **3** is to assign a degree to the membership relation. Since the set where this *grading* is done is finite, it is possible to give an interpretation or understanding of the *degree* of membership of each element of the universe U in a given partial set $A \in \mathbf{3}^U$, as follows:

- $A_{\oplus}$ is the set of elements of U that are known to belong to the set A ;
- $A_{\ominus}$ is the set of elements of U that are known *not* to belong to the set A ;
- $A_{\odot}$ is the set of elements of U for which it is indeterminate whether they belong or *do not* belong to the set A .

Remark 3.2 Let $U \neq \emptyset$. We call a *partial n -ary relation* over U any partial set in $\mathbf{3}^{(U^n)}$. Mikenberg et al. in [16] present a generalization of the Tarskian concept of truth in terms of partial structures and quasi-truth. Their aim was to define a framework where it is possible to manipulate information of partial character coming from scientific domains. The notion of partial structure is introduced based on a suitable notion of *partial relation*. According to Mikenberg et al., a *partial relation* is a relation that is not necessarily defined for every n -tuple of objects from its domain. More precisely, a partial n -ary relation over U is an ordered triple $\langle R_{\oplus}, R_{\ominus}, R_{\odot} \rangle$, where $R_{\oplus}$, $R_{\ominus}$, and $R_{\odot}$ are mutually disjoint and $R_{\oplus} \cup R_{\ominus} \cup R_{\odot} = U^n$. It follows that the notion of partial relation introduced in [16] coincides with our notion of partial relation or **3**-set over U^n .

If we equip **3** with the operations of the matrix $\mathcal{M}_{\mathbf{LFI1}}$, the partial sets naturally inherit these operations, thereby obtaining an algebraic structure. That is, if $A, B \in \mathbf{3}^U$ and $* \in \Theta$, we define

$$\begin{aligned} (*A)(x) &:= *(A(x)) & * &\in \{\neg, \circ\} \\ (A * B)(x) &:= A(x) * B(x) & * &\in \{\wedge, \vee, \rightarrow\} \end{aligned}$$

Proposition 3.3 Let $A = (A_{\oplus}, A_{\ominus}, A_{\odot})$ and $B = (B_{\oplus}, B_{\ominus}, B_{\odot})$ be two **LFI1**-sets. Then

- (a) $\neg A = (A_{\ominus}, A_{\oplus}, A_{\odot})$
- (b) $\circ A = (A_{\oplus} \cup A_{\ominus}, A_{\odot}, \emptyset)$
- (c) $A \wedge B = (A_{\oplus} \cap B_{\oplus}, A_{\ominus} \cup B_{\ominus}, (A_{\oplus} \cap B_{\odot}) \cup (A_{\odot} \cap B_{\oplus}) \cup (A_{\odot} \cap B_{\odot}))$
- (d) $A \vee B = (A_{\oplus} \cup B_{\oplus}, A_{\ominus} \cap B_{\ominus}, (A_{\odot} \cap B_{\odot}) \cup (A_{\ominus} \cap B_{\odot}) \cup (A_{\odot} \cap B_{\oplus}))$
- (e) $A \rightarrow B = (A_{\ominus} \cup B_{\oplus}, (A_{\oplus} \cup A_{\odot}) \cap B_{\ominus}, (A_{\oplus} \cup A_{\odot}) \cap B_{\odot})$

Proof. We only prove (e) and left the rest to the reader.

$$\begin{aligned} x \in (A \rightarrow B)_{\oplus} &\text{ IFF } A(x) \rightarrow B(x) = 1 \\ &\text{ IFF } A(x) = 0 \text{ OR } B(x) = 1 \\ &\text{ IFF } x \in A_{\ominus} \cup A_{\oplus} \end{aligned}$$

$$\begin{aligned} x \in (A \rightarrow B)_{\ominus} &\text{ IFF } A(x) \rightarrow B(x) = 0 \\ &\text{ IFF } A(x) \in \{1, 1/2\} \text{ AND } B(x) = 0 \\ &\text{ IFF } x \in (A_{\oplus} \cup A_{\odot}) \cap B_{\ominus} \end{aligned}$$

$$\begin{aligned} x \in (A \rightarrow B)_{\odot} &\text{ IFF } A(x) \rightarrow B(x) = 1/2 \\ &\text{ IFF } A(x) \in \{1, 1/2\} \text{ AND } B(x) = 1/2 \\ &\text{ IFF } x \in (A_{\oplus} \cup A_{\odot}) \cap B_{\oplus} \end{aligned}$$

■

Recall that a first-order signature $\Sigma = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$ is composed of the following elements: a set $\mathcal{P} = \bigcup_{n \in \mathbb{N}} \mathcal{P}_n$ such that, for each $n \geq 1$, $\mathcal{P}_n$ is a set of predicate symbols of arity n ; a set $\mathcal{F} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$ such that, for each $n \geq 1$, $\mathcal{F}_n$ is a set of function symbols of arity n ; and a set $\mathcal{C}$ of individual constants. Let $\mathcal{L}(\Sigma)$ be the first-order language defined as usual from the connectives $\wedge, \vee, \rightarrow, \neg$,

o, the quantifiers $\forall, \exists$, a denumerable set $\mathcal{V}$ of individual variables and a given first-order signature Σ . We denote by $\text{SENT}(\Sigma)$ the set of sentences (formulas without free-variables) over the signature Σ . If φ is a formula, x is a variable and t is a term, then $\varphi(t/x)$ will denote the formula that results from φ by replacing simultaneously all the free occurrences of the variable x by t . We denote by $\text{TER}(\Sigma)$ the set of all Σ -terms.

Definition 3.4 Given $\Sigma = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$, a Σ -partial structure is a pair $\mathfrak{A} = \langle A, \mathcal{I} \rangle$, where A is a non-empty set and $\mathcal{I}$ is a function defined on Σ such that:

- If $R \in \mathcal{P}_n$, then $\mathcal{I}(R) = R^{\mathfrak{A}}$, where $R^{\mathfrak{A}}$ is a partial n -ary relation on A , denoted as
$$R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle.$$
- If $f \in \mathcal{F}_n$, then $\mathcal{I}(f) = f^{\mathfrak{A}}$, where $f^{\mathfrak{A}} : A^n \rightarrow A$ is a function.
- If $c \in \mathcal{C}$, then $\mathcal{I}(c) = c^{\mathfrak{A}}$ is an element of A .

The set A is called the *domain*, *universe*, or *underlying set* of $\mathfrak{A}$ and we note $|\mathfrak{A}| = A$.

Definition 3.5 Let $\mathfrak{A} = \langle A, \mathcal{I} \rangle$ and $\mathfrak{B} = \langle B, \mathcal{J} \rangle$ be two Σ -partial structures. We say that $\mathfrak{A}$ is a *partial substructure* of $\mathfrak{B}$, denoted by $\mathfrak{A} \subseteq \mathfrak{B}$ if $A \subseteq B$ and the following conditions hold:

1. For every $P \in \mathcal{P}_n$ and $\# \in \{\oplus, \ominus, \odot\}$, we have $P_{\#}^{\mathfrak{A}} = P_{\#}^{\mathfrak{B}} \cap A^n$.
2. For every $f \in \mathcal{F}_n$, $f^{\mathfrak{A}} = f^{\mathfrak{B}}|_{A^n}$ and $f^{\mathfrak{B}}[A^n] \subseteq A$.
3. For every $c \in \mathcal{C}$, $c^{\mathfrak{A}} = c^{\mathfrak{B}}$.

Remark 3.6 As in the usual definition of structure, it is necessary that condition (2) above guarantees that $f^{\mathfrak{B}}(a_1, \dots, a_n) \in A$ for all $a_1, \dots, a_n \in A$. Likewise, $c^{\mathfrak{B}} \in A$, for every $c \in \mathcal{C}$.

Definition 3.7 (Partial Structures with Equality) Suppose the signature Σ contains a binary predicate symbol $E \in \mathcal{P}_2$. If $\mathfrak{A}$ is a Σ -partial structure such that

$$E_{\oplus}^{\mathfrak{A}} \cup E_{\odot}^{\mathfrak{A}} = \{(x, x) \mid x \in A\},$$

we say that $\langle E \rangle$ is a *partial equality* and that $\mathfrak{A}$ is a *normal* or *partial structure with equality*.

It should be noted that, unlike equality in Tarskian structures, there is no unique way to define a partial equality on a given universe (see [12]).

Definition 3.8 Let Σ and Σ' be two signatures such that $\Sigma \subseteq \Sigma'$. If $\mathfrak{A} = \langle A, \mathcal{I} \rangle$ is a Σ -partial structure, then clearly we can extend $\mathcal{I}$ to a function $\mathcal{I}'$ defined on Σ' . In this case, $\mathfrak{A}' = \langle A, \mathcal{I}' \rangle$ is a Σ' -structure called an *expansion* of $\mathfrak{A}$, and $\mathfrak{A}$ is the *reduct* of $\mathfrak{A}'$ to Σ .

Given a Σ' -partial structure $\mathfrak{A}' = \langle A, \mathcal{I}' \rangle$ and $\Sigma \subseteq \Sigma'$, we can restrict $\mathcal{I}'$ to Σ obtaining a function $\mathcal{I}$ defined on Σ . Then, $\mathfrak{A} = \langle A, \mathcal{I} \rangle$ is a Σ -partial structure. Let $\mathfrak{A}$ be a Σ -partial structure. We can add the elements of A to the signature as new constant symbols, obtaining a new signature

$$\Sigma_A := \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \cup A \rangle.$$

We denote by

$$\mathfrak{A}_A \quad \text{or} \quad \langle \mathfrak{A}, A \rangle \quad \text{or} \quad \langle \mathfrak{A}, a \rangle_{a \in A}$$

the expansion of $\mathfrak{A}$ to a Σ_A -partial structure satisfying

$$a^{\langle \mathfrak{A}, A \rangle} = a \quad \text{for every } a \in A.$$

It is not always necessary to consider the full expansion by the set A . If $X \subseteq A$, define

$$\Sigma_X := \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \cup \{a : a \in X\} \rangle,$$

and

$$\mathfrak{A}_X \quad \text{or} \quad \langle \mathfrak{A}, X \rangle \quad \text{or} \quad \langle \mathfrak{A}, a \rangle_{a \in X}$$

is the expansion of $\mathfrak{A}$ to a Σ_X -structure such that $a^{\mathfrak{A}_X} = a$ for every $a \in X$. When $X = \{\mathbf{a}\}$ is a singleton, it is customary to denote

$$(A; \mathbf{a})$$

this expansion. If $\mathfrak{A}$ and $\mathfrak{B}$ are Σ -partial structures, $X \subseteq A$, and

$$f : X \rightarrow B,$$

then $\langle \mathfrak{B}, f(a) \rangle_{a \in X}$ denotes the expansion of $\mathfrak{B}$ to a Σ_X -structure interpreting each constant/parameter a as $f(a)$.

Definition 3.9 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two Σ -partial structures. A *morphism* $h : \mathfrak{A} \rightarrow \mathfrak{B}$ is given by a function $h : A \rightarrow B$ such that:

1. If $P \in \mathcal{P}_n$ and $(a_1, \dots, a_n) \in A^n$, then

$$(a_1, \dots, a_n) \in P_{\oplus}^{\mathfrak{A}} \implies (h(a_1), \dots, h(a_n)) \in P_{\oplus}^{\mathfrak{B}},$$

$$(a_1, \dots, a_n) \in P_{\odot}^{\mathfrak{A}} \implies (h(a_1), \dots, h(a_n)) \in P_{\odot}^{\mathfrak{B}}.$$

2. If $f \in \mathcal{F}_n$ and $(a_1, \dots, a_n) \in A^n$, then

$$h(f^{\mathfrak{A}}(a_1, \dots, a_n)) = f^{\mathfrak{B}}(h(a_1), \dots, h(a_n)).$$

3. If $c \in \mathcal{C}$, then

$$h(c^{\mathfrak{A}}) = c^{\mathfrak{B}}.$$

Definition 3.10 A morphism $h : \mathfrak{A} \rightarrow \mathfrak{B}$ is called an *embedding* if the map $h : A \rightarrow B$ is injective and for every $P \in \mathcal{P}_n$ and $(a_1, \dots, a_n) \in A^n$,

$$(a_1, \dots, a_n) \in P_{\ominus}^{\mathfrak{A}} \implies (h(a_1), \dots, h(a_n)) \in P_{\ominus}^{\mathfrak{B}}.$$

Theorem 3.11 If $h : \mathfrak{A} \rightarrow \mathfrak{B}$ is an embedding, then for every $P \in \mathcal{P}_n$ and $(a_1, \dots, a_n) \in A^n$ the following equivalences hold:

- (1) $(a_1, \dots, a_n) \in P_{\oplus}^{\mathfrak{A}} \iff (h(a_1), \dots, h(a_n)) \in P_{\oplus}^{\mathfrak{B}},$
- (2) $(a_1, \dots, a_n) \in P_{\odot}^{\mathfrak{A}} \iff (h(a_1), \dots, h(a_n)) \in P_{\odot}^{\mathfrak{B}},$
- (3) $(a_1, \dots, a_n) \in P_{\ominus}^{\mathfrak{A}} \iff (h(a_1), \dots, h(a_n)) \in P_{\ominus}^{\mathfrak{B}}.$

Proof. (1). Let $(a_1, \dots, a_n) \in A^n$ be such that $(h(a_1), \dots, h(a_n)) \in P_{\oplus}^{\mathfrak{B}}$. Since $(h(a_1), \dots, h(a_n)) \notin P_{\ominus}^{\mathfrak{B}} \cup P_{\odot}^{\mathfrak{B}}$, it follows from the definition of embedding (Definitions 3.9(1) and 3.10) that $(a_1, \dots, a_n) \notin P_{\ominus}^{\mathfrak{A}} \cup P_{\odot}^{\mathfrak{A}}$. Hence, $(a_1, \dots, a_n) \in P_{\oplus}^{\mathfrak{A}}$. The proofs of (2) and (3) are analogous. ■

Definition 3.12 An embedding $h : \mathfrak{A} \rightarrow \mathfrak{B}$ is an *isomorphism* if h is bijective. In this case, we say that $\mathfrak{A}$ and $\mathfrak{B}$ are *isomorphic* and write

$$\mathfrak{A} \cong \mathfrak{B}.$$

Definition 3.13 Let $\mathcal{L}(\Sigma)$ be a first-order language and let $\mathfrak{A}$ be a partial Σ -structure. An *assignment* in $\mathfrak{A}$ is any function

$$s : \mathcal{V} \rightarrow |\mathfrak{A}|.$$

We denote by

$$S(\mathfrak{A})$$

the set of all assignments in $\mathfrak{A}$, that is, $S(\mathfrak{A}) = A^{\mathcal{V}}$. Given an assignment s in $\mathfrak{A}$, the **value** of a term t in $\mathfrak{A}$ under the assignment s , denoted by

$$t^{\mathfrak{A}}[s],$$

is an element of A defined recursively as follows:

If t is $v \in \mathcal{V}$, then $t^{\mathfrak{A}}[s] := s(v)$.

If t is $c \in \mathcal{C}$, then $t^{\mathfrak{A}}[s] := c^{\mathfrak{A}}$.

If t is $f(t_1, \dots, t_n)$, where $f \in \mathcal{F}_n$ and $t_1, \dots, t_n$ are terms, then $t^{\mathfrak{A}}[s] := f^{\mathfrak{A}}(t_1^{\mathfrak{A}}[s], \dots, t_n^{\mathfrak{A}}[s])$.

Given an assignment s in $\mathfrak{A}$, a variable $x \in \mathcal{V}$, and an element $a \in A$, we define the assignment

$$s_x^a : \mathcal{V} \rightarrow A$$

by

$$s_x^a(y) = \begin{cases} s(y) & \text{if } y \neq x \\ a & \text{if } y = x \end{cases}$$

Definition 3.14 For each variable $x \in \mathcal{V}$ of the first-order language $\mathcal{L}(\Sigma)$, we define the following operations:

$$\begin{array}{ccc} \forall x : & \mathbf{2}^{S(\mathfrak{A})} & \rightarrow & \mathbf{2}^{S(\mathfrak{A})} \\ & X & \mapsto & \forall x(X) \end{array} \quad \text{and} \quad \begin{array}{ccc} \exists x : & \mathbf{2}^{S(\mathfrak{A})} & \rightarrow & \mathbf{2}^{S(\mathfrak{A})} \\ & X & \mapsto & \exists x(X) \end{array}$$

defined by:

$$\begin{aligned} s \in \forall x(X) & \quad \text{if and only if} \quad s_x^a \in X \text{ for all } a \in A, \\ s \in \exists x(X) & \quad \text{if and only if} \quad s_x^a \in X \text{ for some } a \in A. \end{aligned}$$

Proposition 3.15 The operations defined in 3.14 satisfy the following properties:

$$\begin{aligned} (\forall 1) \quad \forall x(X) &\subset X. & (\exists 1) \quad X &\subseteq \exists x(X). \\ (\forall 2) \quad \forall x(\forall x(X)) &= \forall x(X). & (\exists 2) \quad \exists x(\exists x(X)) &= \exists x(X). \\ (\forall 3) \quad \text{If } x \neq y, \text{ then } \forall x(\forall y(X)) &= \forall y(\forall x(X)). & (\exists 3) \quad \text{If } x \neq y, \text{ then } \exists x(\exists y(X)) &= \exists y(\exists x(X)). \\ (\forall 4) \quad \text{If } X \subset Y, \text{ then } \forall x(X) &\subset \forall x(Y). & (\exists 4) \quad \text{If } X \subseteq Y, \text{ then } \exists x(X) &\subset \exists x(Y). \\ (\forall 5) \quad \forall x(X \cap Y) &= \forall x(X) \cap \forall x(Y). & (\exists 5) \quad \exists x(X \cup Y) &= \exists x(X) \cup \exists x(Y). \end{aligned}$$

Proof. From Definition 3.14. ■

Definition 3.16¹ An **QLFI1-structure** is a pair $\langle \mathfrak{A}, \|\cdot\|^\mathfrak{A} \rangle$ where $\mathfrak{A}$ is a partial structure and the map $\|\cdot\|^\mathfrak{A} : \mathcal{L}(\Sigma) \rightarrow 3^{S(\mathfrak{A})}$ is recursively defined, where $\langle 3^{S(\mathfrak{A})}, \wedge, \vee, \rightarrow, \neg, \circ \rangle$ is an algebra of **LFII**-sets (see Proposition 3.3), as follows:

- $\|P(\bar{t})\|^\mathfrak{A} := \langle \|P(\bar{t})\|_\oplus^\mathfrak{A}, \|P(\bar{t})\|_\ominus^\mathfrak{A}, \|P(\bar{t})\|_\odot^\mathfrak{A} \rangle$ where

$$\begin{aligned} s \in \|P(t_1, \dots, t_n)\|_\oplus^\mathfrak{A} & \quad \text{iff} \quad (t_1^\mathfrak{A}[s], \dots, t_n^\mathfrak{A}[s]) \in P_\oplus^\mathfrak{A}, \\ s \in \|P(t_1, \dots, t_n)\|_\ominus^\mathfrak{A} & \quad \text{iff} \quad (t_1^\mathfrak{A}[s], \dots, t_n^\mathfrak{A}[s]) \in P_\ominus^\mathfrak{A}, \\ s \in \|P(t_1, \dots, t_n)\|_\odot^\mathfrak{A} & \quad \text{iff} \quad (t_1^\mathfrak{A}[s], \dots, t_n^\mathfrak{A}[s]) \in P_\odot^\mathfrak{A}. \end{aligned}$$
- $\|\neg\varphi\|^\mathfrak{A} := \neg\|\varphi\|^\mathfrak{A},$
- $\|\circ\varphi\|^\mathfrak{A} := \circ\|\varphi\|^\mathfrak{A},$
- $\|\varphi \wedge \psi\|^\mathfrak{A} := \|\varphi\|^\mathfrak{A} \wedge \|\psi\|^\mathfrak{A},$
- $\|\varphi \vee \psi\|^\mathfrak{A} := \|\varphi\|^\mathfrak{A} \vee \|\psi\|^\mathfrak{A},$
- $\|\varphi \rightarrow \psi\|^\mathfrak{A} := \|\varphi\|^\mathfrak{A} \rightarrow \|\psi\|^\mathfrak{A},$
- $\|\forall x\varphi\|^\mathfrak{A} := \langle \forall x\|\varphi\|_\oplus^\mathfrak{A}, \exists x\|\varphi\|_\ominus^\mathfrak{A}, \exists x\|\varphi\|_\odot^\mathfrak{A} - \exists x\|\varphi\|_\ominus^\mathfrak{A} \rangle,$
- $\|\exists x\varphi\|^\mathfrak{A} := \langle \exists x\|\varphi\|_\oplus^\mathfrak{A}, \forall x\|\varphi\|_\ominus^\mathfrak{A}, \exists x\|\varphi\|_\odot^\mathfrak{A} - \exists x\|\varphi\|_\oplus^\mathfrak{A} \rangle.$

Remark 3.17 The map $\|\cdot\|^\mathfrak{A}$ is well defined; we will prove this for the formulas $\forall x\varphi$ and $\exists x\varphi$. For the other formulas, this follows trivially from 3.3.

Proposition 3.18 $\|\forall x\varphi\|$ is a partial set.

¹cf. [8], paragraph 1.9, p. 7.

Proof. Suppose that $s \in \forall x \|\varphi\|_{\oplus} \cap \exists x \|\varphi\|_{\ominus}$, then $s_x^a \in \|\varphi\|_{\oplus}$ for every $a \in A$, and $s_x^b \in \|\varphi\|_{\ominus}$ for some $b \in A$. It follows that $s_x^b \in \|\varphi\|_{\oplus} \cap \|\varphi\|_{\ominus}$, which is a contradiction. Now, since

$$\begin{aligned} s \in \exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\ominus} & \text{ ONLY IF } \exists a, s_x^a \in \|\varphi\|_{\odot} \\ & \text{ ONLY IF } s \notin \forall x \|\varphi\|_{\oplus} \end{aligned}$$

we obtain that

$$\forall x \|\varphi\|_{\oplus} \cap (\exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\ominus}) = \emptyset.$$

On the other hand,

$$\begin{aligned} s \notin \forall x \|\varphi\|_{\oplus} \cup \exists x \|\varphi\|_{\ominus} & \text{ IMPLIES } s \notin \forall x \|\varphi\|_{\oplus} \text{ AND } s \notin \exists x \|\varphi\|_{\ominus} \\ & \text{ IMPLIES } \exists a, s_x^a \notin \|\varphi\|_{\oplus} \cup \|\varphi\|_{\ominus} \text{ AND } s \notin \exists x \|\varphi\|_{\ominus} \\ & \text{ IMPLIES } \exists a, s_x^a \in \|\varphi\|_{\odot} \text{ AND } s \notin \exists x \|\varphi\|_{\ominus} \\ & \text{ IMPLIES } s \in \exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\ominus} \text{ AND } s \notin \exists x \|\varphi\|_{\ominus} \end{aligned}$$

Thus,

$$\exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\ominus} = S(\mathfrak{A}) - (\forall x \|\varphi\|_{\oplus} \cup \exists x \|\varphi\|_{\ominus}).$$

■

Proposition 3.19 $\|\exists x \varphi\|$ is a partial set.

Proof. Suppose that $s \in \exists x \|\varphi\|_{\oplus} \cap \forall x \|\varphi\|_{\ominus}$. Then, there exists $b \in A$ such that $s_b^b \in \|\varphi\|_{\oplus} \cap \|\varphi\|_{\ominus}$. Hence, $\|\exists x \varphi\|_{\oplus} \cap \|\forall x \varphi\|_{\ominus} = \emptyset$. Moreover,

$$\begin{aligned} s \notin \|\exists x \varphi\|_{\oplus} \cup \|\forall x \varphi\|_{\ominus} & \text{ ONLY IF } s \notin \|\exists x \varphi\|_{\oplus} \text{ AND } s \notin \|\forall x \varphi\|_{\ominus} \\ & \text{ ONLY IF } \exists a, s_x^a \notin \|\varphi\|_{\oplus} \cup \|\varphi\|_{\ominus} \\ & \text{ ONLY IF } \exists a, s_x^a \in \|\varphi\|_{\odot} \\ & \text{ ONLY IF } s \in \exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\oplus} \end{aligned}$$

and

$$\begin{aligned} s \in \exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\oplus} & \text{ ONLY IF } \exists a, s_x^a \in \|\varphi\|_{\odot} \\ & \text{ ONLY IF } s \notin \forall x \|\varphi\|_{\oplus} \end{aligned}$$

Thus,

$$\exists x \|\varphi\|_{\odot} - \exists x \|\varphi\|_{\oplus} = S(\mathfrak{A}) - (\|\exists x \varphi\|_{\oplus} \cup \|\forall x \varphi\|_{\ominus}).$$

■

Remark. In this way, given a partial structure $\mathfrak{A}$, we associate to each formula φ an ordered triple or partial set over $S(\mathfrak{A})$:

$$\|\varphi\| = \langle \|\varphi\|_{\oplus}, \|\varphi\|_{\ominus}, \|\varphi\|_{\odot} \rangle$$

with the following intuitive interpretation for $s \in \|\varphi\|_{\#}^{\mathfrak{A}}$, where $\# \in \{\oplus, \odot, \ominus\}$:

$\|\varphi\|_{\oplus}$ is the set of valuations in $\mathfrak{A}$ where φ is interpreted as **true**,

$\|\varphi\|_{\ominus}$ is the set of valuations where φ is interpreted as **false**,

$\|\varphi\|_{\odot}$ is the set of valuations where φ is interpreted as **undetermined** or **inconsistent**.

Definition 3.20 Let $\langle \mathfrak{A}, \|\cdot\| \rangle$ be an **QLFI1**-structure and φ a formula. We say that $s \in S(\mathfrak{A})$ satisfies φ in $\mathfrak{A}$, written

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s],$$

if $s \in \|\varphi\|_{\oplus} \cup \|\varphi\|_{\odot}$.

Proposition 3.21 Let $\mathfrak{A}$ be an **QLFI1**-structure and $s \in S(\mathfrak{A})$. Then:

- (1) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} P(t_1, \dots, t_n)[s]$ if and only if $(t_1, \dots, t_n) \in \|P\|_{\oplus} \cup \|P\|_{\odot}$.
- (2) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \neg\varphi[s]$ if and only if $s \in \|\varphi\|_{\ominus} \cup \|\varphi\|_{\odot}$.
- (3) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \circ\varphi[s]$ if and only if $s \in \|\varphi\|_{\oplus} \cup \|\varphi\|_{\ominus}$.
- (4) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \wedge \psi)[s]$ if and only if $\mathfrak{A} \Vdash \varphi[s]$ and $\mathfrak{A} \Vdash \psi[s]$.
- (5) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \vee \psi)[s]$ if and only if $\mathfrak{A} \Vdash \varphi[s]$ or $\mathfrak{A} \Vdash \psi[s]$.
- (6) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \rightarrow \psi)[s]$ if and only if $\mathfrak{A} \Vdash \varphi[s]$ implies $\mathfrak{A} \Vdash \psi[s]$.
- (7) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\exists x \varphi)[s]$ if and only if there exists $a \in A$ such that $\mathfrak{A} \Vdash \varphi[s_x^a]$.
- (8) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\forall x \varphi)[s]$ if and only if for every $a \in A$, $\mathfrak{A} \Vdash \varphi[s_x^a]$.

Proof. We only prove (4) – (8).
“(4)”.

$$\begin{aligned} \mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \wedge \psi)[s] & \text{ IFF } s \in \|\varphi \wedge \psi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi \wedge \psi\|_{\odot}^{\mathfrak{A}} \\ & \text{ IFF } s \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}) \cup ((\|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}}) \cup (\|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}) \cup (\|\varphi\|_{\odot}^{\mathfrak{A}} \cap \|\psi\|_{\odot}^{\mathfrak{A}})) \\ & \text{ IFF } s \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \cap (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\ & \text{ IFF } s \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \text{ and } s \in (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\ & \text{ IFF } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s] \text{ and } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \psi[s]. \end{aligned}$$

“(5)”.

$$\begin{aligned} \mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \vee \psi)[s] & \text{ IFF } s \notin \|\varphi \vee \psi\|_{\ominus}^{\mathfrak{A}} \\ & \text{ IFF } s \notin \|\varphi\|_{\ominus}^{\mathfrak{A}} \cap \|\psi\|_{\ominus}^{\mathfrak{A}} \\ & \text{ IFF } s \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \cup (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\ & \text{ IFF } s \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \text{ or } s \in (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\ & \text{ IFF } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s] \text{ or } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \psi[s]. \end{aligned}$$

“(6)”.

$$\begin{aligned}
\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \rightarrow \psi)[s] & \text{ IFF } s \notin \|\varphi \rightarrow \psi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF } s \notin (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \cap \|\psi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF } s \in \|\varphi\|_{\ominus}^{\mathfrak{A}} \cup (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\
& \text{ IFF } s \in \|\varphi\|_{\ominus}^{\mathfrak{A}} \text{ OR } s \in (\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}) \\
& \text{ IFF } \mathfrak{A} \not\Vdash_{\mathbf{QLFI1}} \varphi[s] \text{ OR } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \psi[s] \\
& \text{ IFF } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s] \text{ IMPLIES } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \psi[s].
\end{aligned}$$

“(7)”.

$$\begin{aligned}
\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\exists x\varphi)[s] & \text{ IFF } s \notin \|\exists x\varphi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF } s \notin \forall x \|\varphi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF there exists } a \in A \text{ such that } s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} \\
& \text{ IFF there exists } a \in A \text{ such that } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s_x^a]
\end{aligned}$$

“(8)”.

$$\begin{aligned}
\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\forall x\varphi)[s] & \text{ IFF } s \notin \|\forall x\varphi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF } s \notin \exists x \|\varphi\|_{\ominus}^{\mathfrak{A}} \\
& \text{ IFF for every } a \in A, \quad s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} \\
& \text{ IFF for every } a \in A, \quad \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s_x^a]
\end{aligned}$$

■

Remark 3.22 From the previous proof we can extract the following characterization of the satisfiability of a quantified formula under a valuation, which we expect may be useful at some point:

$$(7') \quad \mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\exists x\varphi)[s] \text{ if and only if } s \in \exists x (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}).$$

$$(8') \quad \mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\forall x\varphi)[s] \text{ if and only if } s \in \forall x (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}).$$

Proposition 3.23² Let φ be a formula, $\mathfrak{A}$ an **QLFI1**-structure, and $s \in S(\mathfrak{A})$. Then:

- (1) $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \wedge \circ\varphi)[s]$.
- (2) $s \in \|\varphi\|_{\ominus}^{\mathfrak{A}}$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\neg\varphi \wedge \circ\varphi)[s]$.
- (3) $s \in \|\varphi\|_{\odot}^{\mathfrak{A}}$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} (\varphi \wedge \neg\varphi)[s]$.
- (4) s satisfies in $\mathfrak{A}$ at most two among

$$\varphi, \neg\varphi, \circ\varphi.$$

²cf. [6], Proposition 3.8.

Proof. The result follows immediately from the fact that for every formula φ we have:

$$\begin{aligned}\|\varphi \wedge \circ\varphi\|^{\mathfrak{A}} &= \langle \|\varphi\|_{\oplus}^{\mathfrak{A}}, \|\varphi\|_{\ominus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}, \emptyset \rangle, \\ \|\neg\varphi \wedge \circ\varphi\|^{\mathfrak{A}} &= \langle \|\varphi\|_{\ominus}^{\mathfrak{A}}, \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}, \emptyset \rangle, \\ \|\varphi \wedge \neg\varphi\|^{\mathfrak{A}} &= \langle \emptyset, \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}, \|\varphi\|_{\odot}^{\mathfrak{A}} \rangle, \\ \|\varphi \wedge \neg\varphi \wedge \circ\varphi\|^{\mathfrak{A}} &= \langle \emptyset, S(\mathfrak{A}), \emptyset \rangle.\end{aligned}$$

■

Definition 3.24 Let $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ be an **QLFI1**-structure and φ a formula.

(i) We say that φ is **QLFI1-true** in $\mathfrak{A}$, and write $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$, if

$$\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}).$$

(ii) We say that φ is **QLFI1-false** in $\mathfrak{A}$ if $\|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A})$.

A formula φ is said to be **QLFI1-valid**, denoted $\Vdash_{\mathbf{QLFI1}} \varphi$, if

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$$

for every **QLFI1**-structure $\mathfrak{A}$.

4 Some elementary properties

In what follows, let $\langle \mathfrak{A}, \|\cdot\|^{\mathfrak{A}} \rangle$ be an **QLFI1**-structure and φ a formula.

Proposition 4.1 For every formula φ , it is not the case that

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi, \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \neg\varphi, \text{ and } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \circ\varphi.$$

hold simultaneously.

Proof. Otherwise, we would have

$$\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}), \quad \|\varphi\|_{\ominus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}), \quad \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}),$$

which leads to a contradiction, as these conditions imply $S(\mathfrak{A}) = \emptyset$. ■

Proposition 4.2 The following are equivalent:

- (i) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \rightarrow \psi$;
- (ii) $\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} \subset \|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}}$;
- (iii) $\|\psi\|_{\ominus}^{\mathfrak{A}} \subset \|\varphi\|_{\ominus}^{\mathfrak{A}}$.

Proof. From Proposition 3.21 and Definition 3.16. ■

Proposition 4.3 If $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$ and $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \rightarrow \psi$, then $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \psi$.

Proof. If $\|\varphi\|_{\ominus}^{\mathfrak{A}} = \emptyset$ and $\|\varphi \rightarrow \psi\|_{\ominus}^{\mathfrak{A}} = (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}}) \cap \|\psi\|_{\ominus}^{\mathfrak{A}} = \emptyset$, then it follows that $\|\psi\|_{\oplus}^{\mathfrak{A}} \cup \|\psi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. ■

Proposition 4.4 $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \forall x \varphi$.

Proof. “Only if.” Let $s \in S(\mathfrak{A})$. Since $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$, it follows that $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s_x^a]$ for every $a \in A$. Hence, by Proposition 3.21 (8), we conclude that $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \forall x \varphi[s]$. “If.” This follows from $\|\varphi\|_{\ominus}^{\mathfrak{A}} \subset \exists x \|\varphi\|_{\ominus}^{\mathfrak{A}}$. ■

Recall that an instance of a propositional formula is a first-order formula obtained by replacing each propositional variable occurring in the propositional formula with a first-order formula, with all occurrences of the same propositional variable replaced by the same first-order formula.

Proposition 4.5 Let $\alpha = \varphi[\beta_1/p_1, \dots, \beta_n/p_n]$ be an instance in $\mathcal{L}(\Theta)$ of a propositional formula $\varphi(p_1, \dots, p_n)$ in $\mathcal{L}$. For every $s \in S(\mathfrak{A})$ we define the valuation v_s over the 3-valued matrix for **LFI1** as follows:

$$v_s(p_i) = \begin{cases} 1 & \text{if } s \in \|\beta_i\|_{\oplus}^{\mathfrak{A}} \\ 1/2 & \text{if } s \in \|\beta_i\|_{\odot}^{\mathfrak{A}} \\ 0 & \text{if } s \in \|\beta_i\|_{\ominus}^{\mathfrak{A}} \end{cases}$$

for $1 \leq i \leq n$, and $v_s(p_i)$ is arbitrary otherwise. Then $\|\alpha\|_{\ominus}^{\mathfrak{A}}(s) = v_s(\varphi)$.

Proof. Straightforward by induction on the complexity of φ . ■

Proposition 4.6 If φ is an instance of an **LFI1**-tautology, then $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi$.

Proof. From Proposition 4.5. ■

Proposition 4.7 ³ Let $t(v_1, \dots, v_n)$ be a term and $\alpha(v_1, \dots, v_n)$ a formula. If s and s' are two assignments in $\mathfrak{A}$ such that $s(v_i) = s'(v_i)$ for all i , $1 \leq i \leq n$, then:

- (1) $t^{\mathfrak{A}}[s] = t^{\mathfrak{A}}[s']$.
- (2) $s \in \|\alpha\|_{\#}^{\mathfrak{A}}$ if and only if $s' \in \|\alpha\|_{\#}^{\mathfrak{A}}$, for every $\# \in \{\oplus, \odot, \ominus\}$.
- (3) $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \alpha[s]$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \alpha[s']$.

Proof. Straightforward, by induction on the length of the term t and the complexity of α . ■

Remark 4.8 If t is a closed term in $\mathcal{L}(\Sigma)$ and $\mathfrak{A}$ is a Σ -structure, then by Proposition 4.7(1), for any assignments $s, r \in S(\mathfrak{A})$, we have $t^{\mathfrak{A}}[r] = t^{\mathfrak{A}}[s]$. We write

$$t^{\mathfrak{A}}$$

to denote the value (the same for every assignment) of t in $\mathfrak{A}$.

³cf. [4], Proposition 1.3.16.

Theorem 4.9 Let $\mathfrak{A}$ be an **LFII**-structure and φ a sentence. Then

$$\|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}).$$

Proof. Let φ be a sentence, and let $s \in S(\mathfrak{A})$. Suppose $s \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$. Then for any $r \in S(\mathfrak{A})$ and any free variable x of φ , we have $s(x) = r(x)$. Thus, $\|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$. Similarly, if $s \in \|\varphi\|_{\ominus}^{\mathfrak{A}}$ then $\|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A})$, and if $s \in \|\varphi\|_{\odot}^{\mathfrak{A}}$ then $\|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. ■

Corolario 4.10 Let $\mathfrak{A}$ be an **QLFI**-structure and φ a sentence. Then

$$\mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi \wedge \circ\varphi \quad \text{OR} \quad \mathfrak{A} \Vdash_{\mathbf{QLFI}} \neg\varphi \wedge \circ\varphi \quad \text{OR} \quad \mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi \wedge \neg\varphi.$$

It follows that a sentence is either **QLFI**-true or **QLFI**-false in a partial Σ -structure, a phenomenon that does not hold for arbitrary formulas. However, a sentence may be **QLFI**-true in some partial Σ -structures and **QLFI**-false in others.

Corolario 4.11 Let $\mathfrak{A}$ be an **QLFI**-structure and φ a sentence. Then

$$\mathfrak{A} \Vdash_{\mathbf{QLFI}} \circ\varphi \quad \text{OR} \quad \mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi \wedge \neg\varphi.$$

We can thus see that the consistency operator $\circ$ indicates which sentences are true or false in an absolute sense, whereas sentences whose validity can be considered either doubtful or a matter of opinion are not associated with this operator.

Proposition 4.12 Let t and u be two terms and $s \in S(\mathfrak{A})$. If t' is obtained from t by replacing every occurrence of a variable x with u , then

$$(t')^{\mathfrak{A}}[s] = t^{\mathfrak{A}} \left[s_x^{u^{\mathfrak{A}}[s]} \right].$$

Proposition 4.13 (Substitution Lemma) Let t be a term free for x in φ . The following hold:

- (0) For every $\# \in \{\oplus, \ominus, \odot\}$, $s \in \|\varphi(t/x)\|_{\#}^{\mathfrak{A}}$ if and only if $s_x^{t^{\mathfrak{A}}[s]} \in \|\varphi\|_{\#}^{\mathfrak{A}}$.
- (1) $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi(t/x)[s]$ if and only if $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi \left[s_x^{t^{\mathfrak{A}}[s]} \right]$.
- (2) If $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \forall x\varphi[s]$, then $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi(t/x)[s]$.
- (3) If $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \varphi(t/x)[s]$, then $\mathfrak{A} \Vdash_{\mathbf{QLFI}} \exists x\varphi[s]$.

Proof. It is routine. ■

Theorem 4.14 If t is a term free for x in φ , then

$$\begin{aligned} &\Vdash_{\mathbf{QLFI}} \forall x\varphi \rightarrow \varphi(t/x), \\ &\Vdash_{\mathbf{QLFI}} \varphi(t/x) \rightarrow \exists x\varphi. \end{aligned}$$

Proof. It is consequence of Proposition 4.13. ■

Theorem 4.15

$$(1) \Vdash_{\mathbf{QLFI1}} \forall x \circ \varphi \rightarrow \circ \exists x \varphi.$$

$$(2) \Vdash_{\mathbf{QLFI1}} \neg \forall x \varphi \leftrightarrow \exists x \neg \varphi.$$

$$(3) \Vdash_{\mathbf{QLFI1}} \neg \exists x \varphi \leftrightarrow \forall x \neg \varphi.$$

Proof. Let $\mathfrak{A}$ be an **QLFI1**-structure.

“(1)”.

$$\begin{aligned} \|\circ \exists x \varphi\|_{\ominus}^{\mathfrak{A}} &= \|\exists x \varphi\|_{\ominus}^{\mathfrak{A}} \\ &= \exists x \|\varphi\|_{\ominus}^{\mathfrak{A}} - \exists x \|\varphi\|_{\oplus}^{\mathfrak{A}} \\ &\subset \exists x \|\varphi\|_{\ominus}^{\mathfrak{A}} \\ &= \exists x \|\circ \varphi\|_{\ominus}^{\mathfrak{A}} \\ &= \|\forall x \circ \varphi\|_{\ominus}^{\mathfrak{A}} \end{aligned}$$

“(2)”.

$$\begin{aligned} \|\neg \forall x \varphi\|_{\ominus}^{\mathfrak{A}} &= \|\forall x \varphi\|_{\oplus}^{\mathfrak{A}} \\ &= \forall x \|\varphi\|_{\oplus}^{\mathfrak{A}} \\ &= \forall x \|\neg \varphi\|_{\ominus}^{\mathfrak{A}} \\ &= \|\exists x \neg \varphi\|_{\ominus}^{\mathfrak{A}} \end{aligned}$$

“(3)”.

$$\begin{aligned} \|\neg \exists x \varphi\|_{\ominus}^{\mathfrak{A}} &= \|\exists x \varphi\|_{\oplus}^{\mathfrak{A}} \\ &= \exists x \|\varphi\|_{\oplus}^{\mathfrak{A}} \\ &= \exists x \|\neg \varphi\|_{\ominus}^{\mathfrak{A}} \\ &= \|\forall x \neg \varphi\|_{\ominus}^{\mathfrak{A}} \end{aligned}$$

Proposition 4.16 Suppose that the variable x does not occur free in φ . Then, for every **QLFI1**-structure $\mathfrak{A}$,

$$\text{if } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \rightarrow \psi \quad \text{then} \quad \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \rightarrow \forall x \psi.$$

Proof. Let $\mathfrak{A}$ be a **QLFI1**-structure. Assume that there exists $s \in S(\mathfrak{A})$ such that

$$s \in \|\varphi \rightarrow \forall x \psi\|_{\ominus}^{\mathfrak{A}} = (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}) \cap \exists x \|\psi\|_{\ominus}^{\mathfrak{A}}.$$

Then, there exists $a \in A$ such that $s_x^a \in \|\psi\|_{\ominus}^{\mathfrak{A}}$. Since x does not occur free in φ , it follows from Proposition 4.7 that $s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}$. Hence,

$$s_x^a \in (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}) \cap \|\psi\|_{\ominus}^{\mathfrak{A}} = \|(\varphi \rightarrow \psi)\|_{\ominus}^{\mathfrak{A}}.$$

Proposition 4.17 Suppose that the variable x does not occur free in ψ . Then, for every **QLFI1**-structure $\mathfrak{A}$,

$$\text{if } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \rightarrow \psi \quad \text{then} \quad \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \exists x \varphi \rightarrow \psi.$$

Proof. Let $\mathfrak{A}$ be a **QLFI1**-structure. Assume that there exists $s \in S(\mathfrak{A})$ such that

$$s \in \|\exists x \varphi \rightarrow \psi\|_{\ominus}^{\mathfrak{A}} = (\|\exists x \varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\ominus}^{\mathfrak{A}}) \cup (\|\exists x \varphi\|_{\ominus}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}).$$

If $s \in \|\exists x \varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\ominus}^{\mathfrak{A}}$, then there exists $a \in A$ such that $s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}}$. Since x does not occur free in ψ , it follows from Proposition 4.7 that $s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}$. Thus,

$$s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cap \|\psi\|_{\ominus}^{\mathfrak{A}} \subset (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}) \cap \|\psi\|_{\ominus}^{\mathfrak{A}} = \|(\varphi \rightarrow \psi)\|_{\ominus}^{\mathfrak{A}}.$$

If $s \in \|\exists x \varphi\|_{\ominus}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}}$, then there exists $a \in A$ such that $s_x^a \in \|\varphi\|_{\ominus}^{\mathfrak{A}}$. Since x does not occur free in ψ , it follows from Proposition 4.7 that $s_x^a \in \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}$. Thus,

$$s_x^a \in \|\varphi\|_{\ominus}^{\mathfrak{A}} \cap \|\psi\|_{\oplus}^{\mathfrak{A}} \subset (\|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\ominus}^{\mathfrak{A}}) \cap \|\psi\|_{\oplus}^{\mathfrak{A}} = \|(\varphi \rightarrow \psi)\|_{\oplus}^{\mathfrak{A}}.$$

■

5 The first order logic **QLFI1**

Given a first-order signature Σ , the logic **QLFI1** (in the language $\mathcal{L}(\Sigma)$) is defined as the Hilbert-style deductive system obtained by extending the **LFI1** logic expressed in the language $\mathcal{L}(\Sigma)$ with the following. As usual in first-order languages, we understand the notions of free variable, free term for a variable, and variant formulas in the standard way.

Axiom Schemas

$$(\mathbf{Ax17}) \quad \varphi[x/t] \rightarrow \exists x \varphi, \quad \text{if } t \text{ is a free term for } x \text{ in } \varphi$$

$$(\mathbf{Ax18}) \quad \forall x \varphi \rightarrow \varphi[x/t], \quad \text{if } t \text{ is a free term for } x \text{ in } \varphi$$

$$(\mathbf{Ax19}) \quad \varphi \rightarrow \psi, \quad \text{whenever } \varphi \text{ is a variant of } \psi$$

$$(\mathbf{Ax20}) \quad \neg \forall x \varphi \leftrightarrow \exists x \neg \varphi$$

$$(\mathbf{Ax21}) \quad \neg \exists x \varphi \leftrightarrow \forall x \neg \varphi$$

Inference Rules

$$(\forall\text{-In}) \quad \frac{\alpha \rightarrow \beta}{\alpha \rightarrow \forall x \beta} \quad \text{if } x \text{ does not occur freely in } \alpha,$$

$$(\exists\text{-In}) \quad \frac{\alpha \rightarrow \beta}{\exists x \alpha \rightarrow \beta} \quad \text{if } x \text{ does not occur freely in } \beta.$$

Theorem 5.1 (Deduction Theorem) If in the deduction $\Gamma, \varphi \vdash_{\mathbf{QLFI1}} \psi$, no application of the rules $(\exists\text{-In})$ and $(\forall\text{-In})$ has quantified variables that are free in φ , then $\Gamma \vdash_{\mathbf{QLFI1}} \varphi \rightarrow \psi$. In particular, this holds if $\varphi \in \text{Sent}(\Sigma)$.

Proof. The proof is analogous to the classical case. ■

Theorem 5.2

- (a) $\varphi \rightarrow \psi \vdash_{\mathbf{QLFI1}} (\neg\psi \wedge \circ\psi) \rightarrow (\neg\varphi \wedge \circ\varphi)$
- (b) $\vdash_{\mathbf{QLFI1}} (\neg\exists x\varphi \wedge \circ\exists x\varphi) \rightarrow (\forall x(\neg\varphi \wedge \circ\varphi))$
- (c) If $\Gamma, \alpha \vdash_{\mathbf{QLFI1}} \gamma$ and $\Gamma, \beta \vdash_{\mathbf{QLFI1}} \gamma$, then $\Gamma, \alpha \vee \beta \vdash_{\mathbf{QLFI1}} \gamma$.
- (d) If $\Gamma \vdash_{\mathbf{QLFI1}} \varphi$, then $\Gamma \vdash_{\mathbf{QLFI1}} \forall x\varphi$.

Proof. Routine. ■

Theorem 5.3 (Soundness) If $\Gamma \Vdash_{\mathbf{QLFI1}} \varphi$, then $\vdash_{\mathbf{QLFI1}} \varphi$.

Proof. It is consequence of the results stated in Section 4. ■

Theorem 5.4 $\vdash_{\mathbf{QLFI1}} \forall x \circ \varphi \rightarrow \circ \exists x \varphi$

Proof. First, we prove that

$$\varphi, \neg\exists x\varphi \vdash \exists x(\varphi \wedge \neg\varphi).$$

Indeed:

(1)	φ	Hypothesis
(2)	$\neg\exists x\varphi$	Hypothesis
(3)	$\forall x\neg\varphi$	(Ax21)
(4)	$\neg\varphi$	(Ax18)
(5)	$\varphi \wedge \neg\varphi$	(1), (4), $\wedge$ -introduction
(6)	$\exists x(\varphi \wedge \neg\varphi)$	(Ax17)

By the Deduction Metatheorem, since the only inference rule used in the proof was Modus Ponens, it follows that

$$\vdash \varphi \rightarrow (\neg\exists x\varphi \rightarrow \exists x(\varphi \wedge \neg\varphi)).$$

Applying $(\exists\text{-Intro})$ we obtain

$$\vdash \exists x\varphi \rightarrow (\neg\exists x\varphi \rightarrow \exists x(\varphi \wedge \neg\varphi)).$$

Hence, by the importation rule (inherited from classical positive logic),

$$\vdash (\exists x\varphi \wedge \neg\exists x\varphi) \rightarrow \exists x(\varphi \wedge \neg\varphi).$$

From rule **?(T9)**, we have

$$\vdash \sim \exists x(\varphi \wedge \neg\varphi) \rightarrow \sim (\exists x\varphi \wedge \neg\exists x\varphi).$$

Then, by 5.2(b), transitivity of implication, and Modus Ponens,

$$\vdash \forall x \sim (\varphi \wedge \neg\varphi) \rightarrow \sim (\exists x \varphi \wedge \neg \exists x \varphi).$$

Finally, since in **QLFI1** the equivalence **?(T10)** is a theorem, we obtain

$$\vdash \forall x \circ \varphi \rightarrow \circ \exists x \varphi.$$

In order to prove completeness, some well-known notions and results from classical first-order logic will be adapted to this framework. Recall that a *theory* is any set of sentences (that is, a set of formulas without free variables). We say that a theory $T \subset \text{Sent}(\Sigma)$ is **QLFI1-consistent** if for each $\varphi \in \text{Sent}(\Sigma)$, at most two of the sentences φ , $\neg\varphi$, $\circ\varphi$ are derivable from T in **QLFI1** (see [15, p. 274]; see also [5], where this notion was applied for the first time).

Proposition 5.5 A theory T is **QLFI1**-inconsistent if and only if $T \vdash_{\text{QLFI1}} \varphi$ for every formula φ .

Proof. If T is **QLFI1**-inconsistent, then there exists a sentence ψ such that $T \vdash_{\text{QLFI1}} \psi$, $T \vdash_{\text{QLFI1}} \neg\psi$, and $T \vdash_{\text{QLFI1}} \circ\psi$. For any sentence φ , axiom (**Ax11**) and modus ponens imply $T \vdash_{\text{QLFI1}} \varphi$.

Proposition 5.6 A theory T is **QLFI1**-inconsistent if and only if there exists a sentence φ such that $T \not\vdash_{\text{QLFI1}} \varphi$.

Proof. This is equivalent to Proposition 5.5.

Proposition 5.7 Let T be a theory and φ a sentence.

- (1) φ is not derivable from T in **QLFI1** if and only if $T \cup \{\neg\varphi, \circ\varphi\}$ is **QLFI1**-consistent.
- (2) $\neg\varphi$ is not derivable from T in **QLFI1** if and only if $T \cup \{\varphi, \circ\varphi\}$ is **QLFI1**-consistent.
- (3) $\circ\varphi$ is not derivable from T in **QLFI1** if and only if $T \cup \{\varphi, \neg\varphi\}$ is **QLFI1**-consistent.

Proof. “(1)” Suppose $T \cup \{\neg\varphi, \circ\varphi\}$ is not **QLFI1**-consistent. By Proposition 5.6, then $T \cup \{\neg\varphi\} \vdash_{\text{QLFI1}} \varphi$, which implies $T \vdash_{\text{QLFI1}} \neg\varphi \rightarrow (\circ\varphi \rightarrow \varphi)$. Since $\vdash_{\text{QLFI1}} (\neg\varphi \rightarrow (\circ\varphi \rightarrow \varphi)) \rightarrow \varphi$, it follows that $T \vdash_{\text{QLFI1}} \varphi$.

“(2)” Assume $T \cup \{\varphi, \circ\varphi\}$ is not **QLFI1**-consistent. Then $T \cup \{\varphi, \circ\varphi\} \vdash_{\text{QLFI1}} \varphi$, leading to $T \vdash_{\text{QLFI1}} \varphi \rightarrow (\circ\varphi \rightarrow \neg\varphi)$. Since $\vdash_{\text{QLFI1}} (\varphi \rightarrow (\circ\varphi \rightarrow \neg\varphi)) \rightarrow \neg\varphi$, then $T \vdash_{\text{QLFI1}} \neg\varphi$.

“(3)” The reasoning is analogous, using the instance of the **LFII**-tautology $\vdash_{\text{QLFI1}} (\varphi \rightarrow (\neg\varphi \rightarrow \circ\varphi)) \rightarrow \circ\varphi$.

A theory T is called **QLFI1-maximal** if it is **QLFI1**-consistent and not properly included in any other **QLFI1**-consistent theory.

Proposition 5.8 Let T be a **QLFI1**-maximal consistent theory, and φ, ψ sentences. The following hold:

- (1) $\varphi \in T$ iff $T \vdash_{\mathbf{QLFI1}} \varphi$.
- (2) $\varphi \notin T$ iff $\neg\varphi \in T$.
- (3) $\neg\varphi \notin T$ iff $\{\varphi, \circ\varphi\} \subset T$.
- (4) $\circ\varphi \notin T$ iff $\{\varphi, \neg\varphi\} \subset T$.
- (5) $\varphi \wedge \psi \in T$ iff $\varphi \in T$ and $\psi \in T$.
- (6) $\varphi \vee \psi \in T$ iff $\varphi \in T$ or $\psi \in T$.
- (7) Either $\varphi \wedge \circ\varphi, \neg\varphi \wedge \circ\varphi$, or $\varphi \wedge \neg\varphi$ is in T .
- (8) $\forall x\varphi(x) \in T$ only if $\varphi(c) \in T$ for all $c \in \mathcal{C}$, provided that $\mathcal{C} \neq \emptyset$

Proof. It is consequence of what was previously stated. ■

Theorem 5.9 (Lindenbaum-Łoś) Every **QLFI1**-consistent theory can be extended to a **QLFI1**-maximal consistent theory.

Proof. Let $T \subset \text{Sent}(\Sigma)$ be **QLFI1**-consistent. Enumerate all sentences as $\{\varphi_0, \varphi_1, \dots\}$ and construct the following chain of theories:

$$T = T_0 \subset T_1 \subset \dots \subset T_n \subset \dots,$$

where, given T_m , define

$$T_{m+1} = \begin{cases} T_m \cup \{\varphi_m, \neg\varphi_m\} & \text{if } T_m \not\vdash_{\mathbf{QLFI1}} \circ\varphi_m, \\ T_m \cup \{\varphi_m, \circ\varphi_m\} & \text{if } T_m \not\vdash_{\mathbf{QLFI1}} \neg\varphi_m, \\ T_m \cup \{\neg\varphi_m, \circ\varphi_m\} & \text{if } T_m \not\vdash_{\mathbf{QLFI1}} \varphi_m. \end{cases}$$

From the fact that T_0 is consistent and Proposition 5.7, we have that each T_k is consistent. Let $J = \bigcup_{n \in \omega} T_n$, then, J is **QLFI1**-consistent. Indeed, if J is not **QLFI1**-consistent, since $\vdash_{\mathbf{QLFI1}}$ is finitary, there would be some inconsistent T_i . To show J is maximal, assume a **QLFI1**-consistent theory K with $J \subsetneq K$. For any $\sigma \in K$, there exists k such that $\sigma = \varphi_k$. Since $\varphi_k \in K$, it must be that $K \not\vdash_{\mathbf{QLFI1}} \circ\varphi_k$ or $K \not\vdash_{\mathbf{QLFI1}} \neg\varphi_k$, which implies $T_k \not\vdash_{\mathbf{QLFI1}} \circ\varphi_k$ or $T_k \not\vdash_{\mathbf{QLFI1}} \neg\varphi_k$, and by the construction of the chain, this leads to $\varphi_k \in J$, hence $J = K$. ■

We say that $T \subset \text{Sent}(\Sigma)$ is a *Henkin theory for QLFI1* if for every formula $\varphi(x)$ with x as its only free variable, there exist terms t_1 and t_2 without variables such that

$$\begin{aligned} T \vdash_{\mathbf{QLFI1}} ((\exists x)\varphi \wedge \circ(\exists x)\varphi) &\rightarrow (\varphi(t_1) \wedge \circ\varphi(t_1)) \\ T \vdash_{\mathbf{QLFI1}} ((\exists x)\varphi \wedge \neg(\exists x)\varphi) &\rightarrow (\varphi(t_2) \wedge \neg\varphi(t_2)) \end{aligned}$$

Lemma 5.10 (Henkin Lemma) If T is a maximal **QLFI1**-consistent Henkin theory for **QLFI1**, then T has an **QLFI1**-model.

Proof. We will construct an **LFI1**-structure $\mathfrak{A}$ over Σ as follows:

- The universe A is the set of all closed terms (i.e., without variables) of the language $\mathcal{L}(\Sigma)$.
- If c is a constant symbol in the signature, $c^{\mathfrak{A}} := c$.
- If f is an n -ary function symbol in the signature, $f^{\mathfrak{A}} : A^n \rightarrow A$ is defined as $f^{\mathfrak{A}}(a_1, \dots, a_n) := f(a_1, \dots, a_n)$.
- If R is an n -ary predicate symbol, then $R^{\mathfrak{A}} = \langle R_{\oplus}^{\mathfrak{A}}, R_{\ominus}^{\mathfrak{A}}, R_{\odot}^{\mathfrak{A}} \rangle$ is defined as follows:

$$\begin{aligned} (a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}} & \quad \text{IFF} \quad R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ (a_1, \dots, a_n) \in R_{\ominus}^{\mathfrak{A}} & \quad \text{IFF} \quad \neg R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ (a_1, \dots, a_n) \in R_{\odot}^{\mathfrak{A}} & \quad \text{IFF} \quad R(a_1, \dots, a_n) \wedge \neg R(a_1, \dots, a_n) \in T \end{aligned}$$

First, we will prove that

$$\begin{aligned} \|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad \varphi \wedge \circ \varphi \in T \\ \|\varphi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad \neg \varphi \wedge \circ \varphi \in T \\ \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad \varphi \wedge \neg \varphi \in T \end{aligned}$$

for every sentence φ . The proof will be carried out by induction on the complexity of the formula. We analyze a few cases.

Atomic case If φ is $R(a_1, \dots, a_n)$, then

$$\begin{aligned} \|R(a_1, \dots, a_n)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad (a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}} \\ & \quad \text{IFF} \quad R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ \|R(a_1, \dots, a_n)\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad (a_1, \dots, a_n) \in R_{\ominus}^{\mathfrak{A}} \\ & \quad \text{IFF} \quad \neg R(a_1, \dots, a_n) \wedge \circ R(a_1, \dots, a_n) \in T \\ \|R(a_1, \dots, a_n)\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad (a_1, \dots, a_n) \in R_{\odot}^{\mathfrak{A}} \\ & \quad \text{IFF} \quad R(a_1, \dots, a_n) \wedge \neg R(a_1, \dots, a_n) \in T \end{aligned}$$

Case 1 If φ is $\neg \psi$, then

Case 1.1

$$\begin{aligned} \|\neg \psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad \|\psi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \\ & \quad \text{IFF} \quad \neg \psi \wedge \circ \psi \in T & \quad \text{By I.H.} \\ & \quad \text{IFF} \quad \neg \psi \wedge \circ \neg \psi \in T & \quad \text{Since } \circ \psi \equiv \circ \neg \psi \end{aligned}$$

Case 1.2

$$\begin{aligned} \|\neg \psi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) & \quad \text{IFF} \quad \|\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\ & \quad \text{IFF} \quad \psi \wedge \circ \psi \in T & \quad \text{By I.H.} \\ & \quad \text{IFF} \quad \neg \neg \psi \wedge \circ \neg \psi \in T & \quad \text{Since } \circ \psi \equiv \circ \neg \psi \end{aligned}$$

Case 1.3

$$\begin{aligned}
& \|\neg\psi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ IFF } \|\psi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \psi \wedge \neg\psi \in T \quad \text{By I.H.} \\
& \text{IFF } \neg\psi \wedge \neg\neg\psi \in T \quad \text{Since } \neg\neg\psi \equiv \psi
\end{aligned}$$

Case 2 If φ is $\circ\psi$, then

Case 2.1.

$$\begin{aligned}
& \|\circ\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \|\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ OR } \|\psi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \psi \wedge \circ\psi \in T \text{ OR } \neg\psi \wedge \circ\psi \in T \quad \text{By I.H.} \\
& \text{IFF } \circ\psi \in T \\
& \text{IFF } \circ\psi \wedge \circ\circ\psi \in T \quad \text{Since } \circ\psi \equiv \circ\psi \wedge \circ\circ\psi
\end{aligned}$$

Case 2.2.

$$\begin{aligned}
& \|\circ\psi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \|\psi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ OR } \|\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \psi \wedge \neg\psi \in T \quad \text{By I.H.} \\
& \text{IFF } \neg\circ\psi \in T \quad \text{Since } \psi \wedge \neg\psi \rightarrow \neg\circ\psi \\
& \text{IFF } \neg\circ\psi \wedge \circ\circ\psi \in T \quad \text{Since } \circ\psi \equiv \circ\psi \wedge \circ\circ\psi
\end{aligned}$$

Case 2.3.

It follows from the fact that for any sentence $\|\psi\|_{\odot}^{\mathfrak{A}} = \emptyset$ and $\circ\psi \wedge \neg\circ\psi$ cannot be a member of a consistent theory.

Case 3 If φ is $\psi \wedge \xi$, then:

Case 3.1.

$$\begin{aligned}
& \|\psi \wedge \xi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \|\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ AND } \|\xi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \psi \wedge \circ\psi \in T \text{ AND } \xi \wedge \circ\xi \in T \\
& \text{IFF } (\psi \wedge \circ\psi) \wedge (\xi \wedge \circ\xi) \in T \\
& \text{IFF } (\psi \wedge \xi) \wedge \circ(\psi \wedge \xi) \in T.
\end{aligned}$$

Case 3.2.

$$\begin{aligned}
& \|\psi \wedge \xi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \|\psi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ OR } \|\xi\|_{\ominus}^{\mathfrak{A}} = S(\mathfrak{A}) \\
& \text{IFF } \neg\psi \wedge \circ\psi \in T \text{ OR } \neg\xi \wedge \circ\xi \in T \quad \text{I.H.} \\
& \text{IFF } (\neg\psi \wedge \circ\psi) \vee (\neg\xi \wedge \circ\xi) \in T \quad \text{Maximal QLF11-consistency} \\
& \text{IFF } \neg(\psi \wedge \xi) \wedge \circ(\psi \wedge \xi) \in T. \quad \text{From } (\neg\alpha \wedge \circ\alpha) \vee (\neg\beta \wedge \circ\beta) \equiv \\
& \quad \neg(\alpha \wedge \beta) \wedge \circ(\alpha \wedge \beta)
\end{aligned}$$

Case 3.3. If $\|\psi \wedge \xi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$, then

$$\|\psi\|_{\odot}^{\mathfrak{A}} \cap \|\xi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\psi\|_{\odot}^{\mathfrak{A}} \cap \|\xi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\psi\|_{\oplus}^{\mathfrak{A}} \cap \|\xi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A});$$

then, by I.H.,

$$\psi \wedge \neg\psi \wedge \xi \wedge \neg\xi \in T \quad \text{OR} \quad \psi \wedge \neg\psi \wedge \xi \wedge \circ\xi \in T \quad \text{OR} \quad \psi \wedge \circ\psi \wedge \xi \wedge \neg\xi \in T;$$

by the **LFII**-tautologies

$$\begin{aligned} ((\alpha \wedge \neg\alpha) \wedge (\beta \wedge \neg\beta)) &\rightarrow ((\alpha \wedge \beta) \wedge \neg(\alpha \wedge \beta)) \\ ((\alpha \wedge \neg\alpha) \wedge (\beta \wedge \circ\beta)) &\rightarrow ((\alpha \wedge \beta) \wedge \neg(\alpha \wedge \beta)) \end{aligned}$$

it follows that $(\psi \wedge \xi) \wedge \neg(\psi \wedge \xi) \in T$. On the other hand, $(\psi \wedge \xi) \wedge \neg(\psi \wedge \xi) \in T$ only if, by maximal **QLFI1**-consistency, $(\psi \wedge \neg\psi) \wedge \xi \in T$ OR $\psi \wedge (\xi \wedge \neg\xi) \in T$. From the equivalence $\alpha \equiv (\alpha \wedge \circ\alpha) \vee (\alpha \wedge \neg\alpha)$ and maximal **QLFI1**-consistency again,

$$\begin{aligned} (\varphi \wedge \circ\varphi) \wedge (\xi \wedge \neg\xi) &\in T \quad \text{OR} \quad (\varphi \wedge \neg\varphi) \wedge (\xi \wedge \neg\xi) \in T \quad \text{OR} \\ (\varphi \wedge \neg\varphi) \wedge (\xi \wedge \circ\xi) &\in T. \end{aligned}$$

By I.H.,

$$\|\psi\|_{\odot}^{\mathfrak{A}} \cap \|\xi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\psi\|_{\odot}^{\mathfrak{A}} \cap \|\xi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \quad \text{OR} \quad \|\psi\|_{\oplus}^{\mathfrak{A}} \cap \|\xi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A});$$

Consequently, $\|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$.

Case 4 If φ is $\exists x\psi(x)$, then:

Case 4.1. If $\exists x\psi \wedge \circ\exists x\psi \in T$, since T is Henkin for **QLFI1**, there exists $w \in A$ such that $\psi(w) \wedge \circ\psi(w) \in T$. By I.H., $\|\psi(w)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$ and consequently $\exists x\|\psi(x)\|_{\oplus}^{\mathfrak{A}} = \|\exists x\psi(x)\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$.

Case 4.2. If $\neg\exists x\psi(x) \wedge \circ\exists x\psi(x) \in T$, by Theorem 5.2(b), we have $\forall x(\neg\psi(x) \wedge \circ\psi(x)) \in T$. Then, by Prop. 5.8(8), $\neg\psi(a) \wedge \circ\psi(a) \in T$ for all $a \in A$. By I.H., $\|\psi(a)\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$ for all $a \in A$. Therefore, $\forall x\|\psi(x)\|_{\odot}^{\mathfrak{A}} = \|\exists x\psi(x)\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$.

Case 4.3. If $\exists x\psi \wedge \neg\exists x\psi \in T$, since T is Henkin for **QLFI1**, there exists $w \in A$ such that $\psi(w) \wedge \neg\psi(w) \in T$. By I.H., it follows that $\|\psi(w)\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$ and therefore $\exists x\|\psi(x)\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A})$. From Prop. 5.8(5) and **(AxEqEx)**, we have $\forall x\neg\psi \in T$, and by Prop. 5.8(8), $\neg\psi(a) \in T$ for all $a \in A$. By the inductive hypothesis, $\|\psi(a)\|_{\oplus}^{\mathfrak{A}} = \emptyset$ for all $a \in A$, from which it follows that $\exists x\|\psi(x)\|_{\oplus}^{\mathfrak{A}} = \emptyset$. Thus, $\|\exists x\psi\|_{\odot}^{\mathfrak{A}} = \exists x\|\psi\|_{\odot}^{\mathfrak{A}} - \exists x\|\psi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A})$.

Finally, let φ be a sentence

$$\begin{aligned} \varphi \in T &\text{ IFF } \varphi \wedge \circ\varphi \in T \text{ OR } \varphi \wedge \neg\varphi \in T \\ &\text{ IFF } \|\varphi\|_{\oplus}^{\mathfrak{A}} = S(\mathfrak{A}) \text{ OR } \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \\ &\text{ IFF } \|\varphi\|_{\oplus}^{\mathfrak{A}} \cup \|\varphi\|_{\odot}^{\mathfrak{A}} = S(\mathfrak{A}) \\ &\text{ IFF } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi \end{aligned}$$

Hence, $\mathfrak{A}$ is a model of T . ■

Proposition 5.11 Every **QLFI1**-consistent theory T in the language $\mathcal{L}(\Sigma)$ can be extended to a **QLFI1**-consistent Henkin theory for **QLFI1** in the language $\mathcal{L}(\Sigma \cup W)$, where W is a set of constant symbols.

Proof. First, let $T \subset \text{Sent}(\Sigma)$ be a **QLFI1**-consistent theory. Let W be a set of new constants, with $|W| = |\mathcal{L}(\Sigma)|$. Let

$$\varphi_0(x_0), \varphi_1(x_1), \varphi_2(x_2), \dots, \varphi_k(x_k), \dots$$

be an enumeration of all formulas with one free variable in the language $\mathcal{L}(\Sigma \cup W)$. We choose a sequence of elements (symbols) in W

$$a_{i0}, a_{i1}, \dots, a_{in}, \dots$$

such that

- each a_{jk} does not occur in the formulas $\varphi_0(x_0), \varphi_1(x_1), \dots, \varphi_{k-1}(x_{k-1})$,
- each a_{jk} is different from each of $a_{i0}, a_{i1}, \dots, a_{ik-1}$.

Consider the sentences

$$(S_k) \quad (\exists x \varphi_k(x) \wedge \circ \exists x \varphi_k(x)) \rightarrow (\varphi(a_{ik}) \wedge \circ \varphi(a_{ik}))$$

Let $T_0 := T$, $T_n := T \cup \{(S_0), (S_1), \dots, (S_{n-1})\}$ for each $n \in \omega$, and let $J := T \cup \{(S_i)\}_{i \in \omega}$. Any proof in J requires only a finite number of (S_i) sentences, and consequently will be a proof in some T_n . Therefore, if every T_n is **QLFI1**-consistent, so is J . We will prove by induction that

all T_n are **QLFI1**-consistent.

Suppose T_n is **QLFI1**-consistent and T_{n+1} is not. By Proposition 5.5, it follows that

$$T_{n+1} \vdash_{\text{QLFI1}} \neg(S_n) \wedge \circ(S_n)$$

and by the Deduction Theorem

$$T_n \vdash_{\text{QLFI1}} (S_n) \rightarrow (\neg(S_n) \wedge \circ(S_n)).$$

Since $\vdash_{\text{QLFI1}} (\alpha \rightarrow (\neg\alpha \wedge \circ\alpha)) \rightarrow (\neg\alpha \wedge \circ\alpha)$, applying MP,

$$T_n \vdash_{\text{QLFI1}} \neg(S_n) \wedge \circ(S_n).$$

Now, from $\vdash_{\text{QLFI1}} (\neg(\alpha \rightarrow \beta) \wedge \circ(\alpha \rightarrow \beta)) \rightarrow (\alpha \wedge (\neg\beta \wedge \circ\beta))$ it follows that

$$T_n \vdash_{\text{QLFI1}} \exists x_n \varphi_n(x_n) \wedge \circ \exists x_n \varphi_n(x_n), \quad T_n \vdash_{\text{QLFI1}} \neg(\varphi_n(a_n) \wedge \circ \varphi_n(a_n)) \wedge \circ (\varphi_n(a_n) \wedge \circ \varphi_n(a_n)).$$

By $\vdash_{\text{QLFI1}} \neg(\alpha \wedge \circ\alpha) \wedge \circ(\alpha \wedge \circ\alpha) \rightarrow \neg\alpha$, we have

$$T_n \vdash_{\text{QLFI1}} \neg\varphi_n(a_n).$$

Since a_n does not occur in $(S_0), \dots, (S_{n-1})$, we choose a variable x_k that does not occur in the proof of $\neg\varphi_n(a_n)$, so that

$$T_n \vdash_{\text{QLFI1}} \neg\varphi(x_k),$$

By Theorem 5.2(d)

$$T_n \vdash_{\text{QLFI1}} \forall x_k \neg\varphi_n(x_k),$$

It follows from $\vdash_{\text{QLFI1}} \forall x_k \neg\varphi_n(x_k) \rightarrow \neg\exists x_k \neg\neg\varphi(x_k)$ that

$$T_n \vdash_{\text{QLFI1}} \neg\exists x_k \varphi_n(x_k).$$

From **(Ax19)** and MP

$$T_n \vdash_{\mathbf{QLFI1}} \neg \exists x_n \varphi_n(x_n).$$

Hence,

$$T_n \vdash_{\mathbf{QLFI1}} \exists x_n \varphi_n(x_n) \wedge \neg \exists x_n \varphi_n(x_n) \wedge \circ \exists x_n \varphi_n(x_n).$$

It follows that all T_n are **QLFI1**-consistent and consequently

J is **QLFI1**-consistent.

Now, consider the sentences

$$(S_k) \quad (\exists x \varphi_k(x) \wedge \neg \exists x \varphi_k(x)) \rightarrow (\varphi(a_{ik}) \wedge \neg \varphi(a_{ik}))$$

Let $J_0 := T$, $J_n := J \cup \{(S_0), (S_1), \dots, (S_{k-1})\}$ for all $n \in \omega$, and let $K := J \cup \{(S_i)\}_{i \in \omega}$. Any proof in K involves only a finite number of (S_i) sentences, and consequently will be a proof in some J_n . Therefore, if every J_n is **QLFI1**-consistent, so is K . We will prove by induction that all J_n are **QLFI1**-consistent. Suppose J_n is consistent and J_{n+1} is not **QLFI1**-consistent. By Proposition 5.5, it follows that

$$J_{n+1} \vdash_{\mathbf{QLFI1}} \neg(S_n) \wedge \circ(S_n)$$

and by the deduction theorem

$$J_n \vdash_{\mathbf{QLFI1}} (S_n) \rightarrow (\neg(S_n) \wedge \circ(S_n))$$

Since $\vdash_{\mathbf{QLFI1}} (\alpha \rightarrow (\neg \alpha \wedge \circ \alpha)) \rightarrow (\neg \alpha \wedge \circ \alpha)$, by MP,

$$J_n \vdash_{\mathbf{QLFI1}} \neg(S_n) \wedge \circ(S_n)$$

Now, from $\vdash_{\mathbf{QLFI1}} (\neg(\alpha \rightarrow \beta) \wedge \circ(\alpha \rightarrow \beta)) \rightarrow (\alpha \wedge (\neg \beta \wedge \circ \beta))$ it follows that

$$J_n \vdash_{\mathbf{QLFI1}} \exists x_n \varphi_n(x_n) \wedge \neg \exists x_n \varphi_n(x_n), \quad J_n \vdash_{\mathbf{QLFI1}} \neg(\varphi_n(a_n) \wedge \neg \varphi_n(a_n)) \wedge \circ(\varphi_n(a_n) \wedge \neg \varphi_n(a_n))$$

By $\vdash_{\mathbf{QLFI1}} \neg(\alpha \wedge \neg \alpha) \wedge \circ(\alpha \wedge \neg \alpha) \rightarrow \circ \alpha$, we have

$$J_n \vdash_{\mathbf{QLFI1}} \circ \varphi_n(a_n)$$

Since a_n does not occur in $(S_0), \dots, (S_{n-1})$, we choose a variable x_k that does not occur in the proof of $\neg \varphi_n(a_n)$, so that

$$J_n \vdash_{\mathbf{QLFI1}} \circ \varphi_n(x_k),$$

Applying Theorem 5.2(d)

$$J_n \vdash_{\mathbf{QLFI1}} \forall x_k \circ \varphi_n(x_k),$$

It follows, from Theorem 5.4 and MP,

$$J_n \vdash_{\mathbf{QLFI1}} \circ \exists x_k \varphi_n(x_k)$$

From Theorem 5.4 and applying MP again

$$J_n \vdash_{\mathbf{QLFI1}} \circ \exists x_n \varphi_n(x_n)$$

Hence,

$$J_n \vdash_{\mathbf{QLFI1}} \exists x_n \varphi_n(x_n) \wedge \neg \exists x_n \varphi_n(x_n) \wedge \circ \exists x_n \varphi_n(x_n)$$

Consequently, K is **QLFI1**-consistent. ■

Theorem 5.12 A theory is **QLFI1**-consistent if and only if it has an **QLFI1**-model.

Proof. “Only if”. Let T be a **QLFI1**-consistent theory in the language $\mathcal{L}(\Sigma)$. By Proposition 5.11, T has a **QLFI1**-consistent Henkin extension K in the language $\mathcal{L}(\Sigma \cup W)$. By Proposition 5.9, K has a maximal **QLFI1**-consistent extension J . The theory J satisfies the conditions of Theorem 5.10, hence J has an **QLFI1**-model $\mathfrak{A}$. The restriction of the **QLFI1**-structure $\mathfrak{A}$ to the signature Σ is a model of T . ■

Theorem 5.13 (Completeness) If $\Gamma \Vdash_{\mathbf{QLFI1}} \varphi$, then $\Gamma \vdash_{\mathbf{QLFI1}} \varphi$.

Proof. Suppose φ is not deducible from Γ , then $\Gamma \cup \{\neg\varphi, \circ\varphi\}$ is **QLFI1**-consistent. Hence, $\Gamma \cup \{\neg\varphi, \circ\varphi\}$ has a model, and consequently it is not the case that $\Gamma \Vdash_{\mathbf{QLFI1}} \varphi$. ■

Theorem 5.14 (Compactness-1) If every finite subtheory of a theory T has an **QLFI1**-model, then T has an **QLFI1**-model.

Proof. If every finite subtheory of T has **QLFI1**-models, by Theorem 5.12, every finite subtheory of T is **QLFI1**-consistent. Necessarily, T must be **QLFI1**-consistent, and consequently it has an **QLFI1**-model. ■

Theorem 5.15 (Compactness-2) Let T be a theory. If $T \Vdash_{\mathbf{QLFI1}} \varphi$, then there exists a finite subtheory $T_0 \subset T$ such that $T_0 \Vdash_{\mathbf{QLFI1}} \varphi$.

Proof. If $T \Vdash_{\mathbf{QLFI1}} \varphi$, it follows from Theorem 5.13 that $T \vdash_{\mathbf{QLFI1}} \varphi$. Then there exists a finite $T_0 \subset T$ such that $T_0 \vdash_{\mathbf{QLFI1}} \varphi$. Again, by Theorem 5.13, we have $T_0 \Vdash_{\mathbf{QLFI1}} \varphi$. ■

6 Some model-theoretical results

In this Section, some important results of classical Model Theory are obtained for this logic, such as Robinson’s joint consistency theorem and amalgamation.

Lemma 6.1 Let $t(x_1, \dots, x_n)$ be a term, $\mathfrak{A}$ a partial structure, and $s \in S(\mathfrak{A})$ such that $s(x_i) = a_i$ for all i ($1 \leq i \leq n$). Then

$$t(x_1, \dots, x_n)^{\mathfrak{A}}[s] = t(a_1, \dots, a_n)^{(\mathfrak{A}, A)}.$$

Proof. Routine. ■

Proposition 6.2 Let $\varphi(x_1, \dots, x_n)$ be a formula, $\mathfrak{A}$ a partial **QLFI1**-structure, and $s \in S(\mathfrak{A})$ such that $s(x_i) = a_i$ for all i ($1 \leq i \leq n$). Then

$$\begin{aligned} s \in \|\varphi(x_1, \dots, x_n)\|_{\oplus}^{\mathfrak{A}} & \text{ if, and only if, } \|\varphi(a_1, \dots, a_n)\|_{\oplus}^{(\mathfrak{A}, A)} = S(\mathfrak{A}); \\ s \in \|\varphi(x_1, \dots, x_n)\|_{\ominus}^{\mathfrak{A}} & \text{ if, and only if, } \|\varphi(a_1, \dots, a_n)\|_{\ominus}^{(\mathfrak{A}, A)} = S(\mathfrak{A}); \\ s \in \|\varphi(x_1, \dots, x_n)\|_{\odot}^{\mathfrak{A}} & \text{ if, and only if, } \|\varphi(a_1, \dots, a_n)\|_{\odot}^{(\mathfrak{A}, A)} = S(\mathfrak{A}). \end{aligned}$$

Proof. Routine ■

Corolario 6.3 Let $\varphi(x_1, \dots, x_n)$ be a formula, $\mathfrak{A}$ an **QLFI1**-structure, and $s \in S(\mathfrak{A})$ such that $s(x_i) = a_i$ for all i ($1 \leq i \leq n$). Then

$$(\mathfrak{A}, A) \Vdash_{\mathbf{QLFI1}} \varphi(a_1, \dots, a_n) \text{ if, and only if, } \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi(x_1, \dots, x_n)[s].$$

Definition 6.4 We call *basic formulas* the formulas

$$\varphi, \quad \neg\varphi, \quad \circ\varphi$$

whenever φ is an atomic formula. We denote by $Basic(\Sigma)$ the set of basic formulas of the language $\mathcal{L}(\Sigma)$.

Definition 6.5 Let $\mathfrak{A}$ be an **QLFI1**-structure over Σ . The *diagram* of $\mathfrak{A}$ is the set of basic sentences of $\mathcal{L}(\Sigma_A)$ that are **QLFI1**-true in $(\mathfrak{A}, h(A))$, i.e.

$$\varphi \in \text{Diag}(\mathfrak{A}) \quad \text{IFF} \quad \varphi \in \text{Sent}(\Sigma_A) \cap \text{Basic}(\Sigma_A) \quad \text{AND} \quad (\mathfrak{A}, h(A)) \Vdash_{\mathbf{QLFI1}} \varphi.$$

Proposition 6.6 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QLFI1**-structures with equality over Σ , and let $h : A \rightarrow B$. The following are equivalent:

- (a) h is an embedding of $\mathfrak{A}$ into $\mathfrak{B}$.
- (b) $(\mathfrak{B}, h(a))_{a \in \mathfrak{A}}$ is a model of $\text{Diag}(\mathfrak{A})$.
- (c) There exists an extension $\mathfrak{C} \supset \mathfrak{A}$ and an isomorphism $g : \mathfrak{A} \rightarrow \mathfrak{B}$ such that $g \supset h$.

Proof. “(a) only if (c)”. We can extend A to a set C and extend the function h to a bijection $g : C \rightarrow B$, taking $C := A \cup (B - h(A))$ and

$$g(x) = \begin{cases} h(x) & \text{if } x \in A \\ x & \text{if } x \in B - h(A) \end{cases}$$

We define the **QLFI1**-structure $\mathfrak{C}$ with domain C as follows:

- If f is an n -ary function symbol,

$$f^{\mathfrak{C}}(c_1, \dots, c_n) = c \quad \text{if, and only if,} \quad f^{\mathfrak{B}}(g(c_1), \dots, g(c_n)) = g(c).$$

- If R is an n -ary predicate symbol, $R^{\mathfrak{C}} = \langle R_{\oplus}^{\mathfrak{C}}, R_{\ominus}^{\mathfrak{C}}, R_{\odot}^{\mathfrak{C}} \rangle$ is determined by

$$\begin{aligned} (c_1, \dots, c_n) \in R_{\oplus}^{\mathfrak{C}} & \quad \text{if, and only if,} \quad (g(c_1), \dots, g(c_n)) \in R_{\oplus}^{\mathfrak{B}} \\ (c_1, \dots, c_n) \in R_{\ominus}^{\mathfrak{C}} & \quad \text{if, and only if,} \quad (g(c_1), \dots, g(c_n)) \in R_{\ominus}^{\mathfrak{B}} \\ (c_1, \dots, c_n) \in R_{\odot}^{\mathfrak{C}} & \quad \text{if, and only if,} \quad (g(c_1), \dots, g(c_n)) \in R_{\odot}^{\mathfrak{B}} \end{aligned}$$

“(c) only if (a)”. Immediate.

“(a) only if (b)”. Let $\varphi \in \text{Diag}(\mathfrak{A})$. There exist $R \in \mathcal{P}_n$ and $a_1, \dots, a_n \in A$ such that φ is one of the following formulas:

$$R(a_1, \dots, a_n), \quad \neg R(a_1, \dots, a_n), \quad \circ R(a_1, \dots, a_n).$$

If φ is $R(a_1, \dots, a_n)$, then

$$\begin{aligned} & (\mathfrak{A}, h(A)) \Vdash_{\mathbf{QLFI1}} R(a_1, \dots, a_n) \\ \text{IFF } & (a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}} \cup R_{\ominus}^{\mathfrak{A}} & 3.21(1) \\ \text{IFF } & (h(a_1), \dots, h(a_n)) \in R_{\oplus}^{\mathfrak{B}} \cup R_{\ominus}^{\mathfrak{B}} & 3.11 \\ \text{IFF } & (\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} R(a_1, \dots, a_n). \end{aligned}$$

If φ is $\neg R(a_1, \dots, a_n)$, then

$$\begin{aligned} & (\mathfrak{A}, h(A)) \Vdash_{\mathbf{QLFI1}} \neg R(a_1, \dots, a_n) \\ \text{IFF } & (a_1, \dots, a_n) \in R_{\ominus}^{\mathfrak{A}} \cup R_{\oplus}^{\mathfrak{A}} & 3.21(2) \\ \text{IFF } & (h(a_1), \dots, h(a_n)) \in R_{\ominus}^{\mathfrak{B}} \cup R_{\oplus}^{\mathfrak{B}} & 3.11 \\ \text{IFF } & (\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} \neg R(a_1, \dots, a_n). \end{aligned}$$

If φ is $\circ R(a_1, \dots, a_n)$, then

$$\begin{aligned} & (\mathfrak{A}, h(A)) \Vdash_{\mathbf{QLFI1}} \circ R(a_1, \dots, a_n) \\ \text{IFF } & (a_1, \dots, a_n) \in R_{\oplus}^{\mathfrak{A}} \cup R_{\ominus}^{\mathfrak{A}} & 3.21(3) \\ \text{IFF } & (h(a_1), \dots, h(a_n)) \in R_{\oplus}^{\mathfrak{B}} \cup R_{\ominus}^{\mathfrak{B}} & 3.11 \\ \text{IFF } & (\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} \circ R(a_1, \dots, a_n). \end{aligned}$$

“(b) only if (a)”. h is injective. Let $a_1, a_2 \in A$.

$a_1 \neq a_2$

$$\begin{aligned} \text{ONLY IF } & \|E(a_1, a_2)\|_{\ominus}^{(\mathfrak{A}, A)} = S(\mathfrak{A}) & 4.9 \\ \text{ONLY IF } & (\mathfrak{A}, A) \Vdash_{\mathbf{QLFI1}} \neg E(a_1, a_2) \wedge \circ E(a_1, a_2) & 3.23(2) \\ \text{ONLY IF } & (\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} \neg E(a_1, a_2) \wedge \circ E(a_1, a_2) & 3.21(4), (b) \\ \text{ONLY IF } & E(a_1, a_2)_{\ominus}^{(\mathfrak{B}, h(a))} = S(\mathfrak{A}) \\ \text{ONLY IF } & h(a_1) \neq h(a_2) \end{aligned}$$

(i). Let $c \in \mathcal{C}$. There exists $a \in A$ such that

$$(*) \quad c^{\mathfrak{A}} = a.$$

Hence $(\mathfrak{A}, A) \Vdash_{\mathbf{QLFI1}} E(c, a)$. By (b), $(\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} E(c, a)$. Thus $c^{(\mathfrak{B}, h(a))} = a^{(\mathfrak{B}, h(a))}$, i.e.

$$(**) \quad c^{\mathfrak{B}} = h(a).$$

Therefore, from $(*)$ and $(**)$, we have $h(c^{\mathfrak{A}}) = c^{\mathfrak{B}}$.

(ii). Let $f \in \mathcal{F}_n$. For every sequence $a_1, \dots, a_n \in A$ there exists $a \in A$ such that

$$(*) \quad f^{\mathfrak{A}}(a_1, \dots, a_n) = a.$$

Thus $(\Vdash_{\mathbf{QLFI1}}, h(A))E(f(a_1, \dots, a_n), a)$. By (b), $(\mathfrak{B}, h(a))_{a \in \mathfrak{A}} \Vdash_{\mathbf{QLFI1}} E(f(a_1, \dots, a_n), a)$. Then,

$$(**) \quad f^{\mathfrak{B}}(h(a_1), \dots, h(a_n)) = h(a).$$

Therefore, from $(*)$ and $(**)$, we obtain $h(f^{\mathfrak{A}}(a_1, \dots, a_n)) = f^{\mathfrak{B}}(h(a_1), \dots, h(a_n))$.
 (iii). Let $P \in \mathcal{P}_n$,

$$\begin{aligned} (a_1, \dots, a_n) &\in P_{\oplus}^{\mathfrak{A}}, \\ \text{ONLY IF } (\mathfrak{A}, A) &\Vdash_{\mathbf{QLFI1}} P(a_1, \dots, a_n) \wedge \circ P(a_1, \dots, a_n) & 3.23 \\ \text{ONLY IF } (\mathfrak{B}, h(a)) &\Vdash_{\mathbf{QLFI1}} P(a_1, \dots, a_n) \wedge \circ P(a_1, \dots, a_n) & (b) \\ \text{ONLY IF } (h(a_1), \dots, h(a_n)) &\in P_{\oplus}^{\mathfrak{B}}. & 3.23 \end{aligned}$$

$$\begin{aligned} (a_1, \dots, a_n) &\in P_{\ominus}^{\mathfrak{A}}, \\ \text{ONLY IF } (\mathfrak{A}, A) &\Vdash_{\mathbf{QLFI1}} P(a_1, \dots, a_n) \wedge \neg P(a_1, \dots, a_n) & 3.23 \\ \text{ONLY IF } (\mathfrak{B}, h(a)) &\Vdash_{\mathbf{QLFI1}} P(a_1, \dots, a_n) \wedge \neg P(a_1, \dots, a_n) & (b) \\ \text{ONLY IF } (h(a_1), \dots, h(a_n)) &\in P_{\ominus}^{\mathfrak{B}}. & 3.23 \end{aligned}$$

$$\begin{aligned} (a_1, \dots, a_n) &\in P_{\odot}^{\mathfrak{A}}, \\ \text{ONLY IF } (\mathfrak{A}, A) &\Vdash_{\mathbf{QLFI1}} \neg P(a_1, \dots, a_n) \wedge \circ P(a_1, \dots, a_n) & 3.23 \\ \text{ONLY IF } (\mathfrak{B}, h(a)) &\Vdash_{\mathbf{QLFI1}} \neg P(a_1, \dots, a_n) \wedge \circ P(a_1, \dots, a_n) & (b) \\ \text{ONLY IF } (h(a_1), \dots, h(a_n)) &\in P_{\odot}^{\mathfrak{B}}. & 3.23 \end{aligned}$$

Therefore, since h is injective, from (i), (ii), and (iii) it follows that h is an embedding (see 3.10) of $\mathfrak{A}$ into $\mathfrak{B}$. ■

Definition 6.7 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QLFI1**-structures over Σ such that $\mathfrak{A} \subseteq \mathfrak{B}$. We say that $\mathfrak{A}$ is an elementary **QLFI1**-substructure of $\mathfrak{B}$, and that $\mathfrak{B}$ is an elementary **QLFI1**-extension of $\mathfrak{A}$, and we write

$$\mathfrak{A} \prec \mathfrak{B},$$

if, for every formula φ and for every $s \in S(\mathfrak{A})$, we have

$$\begin{aligned} s \in \|\varphi\|_{\oplus}^{\mathfrak{A}} &\text{ if and only if } s \in \|\varphi\|_{\oplus}^{\mathfrak{B}} \\ s \in \|\varphi\|_{\ominus}^{\mathfrak{A}} &\text{ if and only if } s \in \|\varphi\|_{\ominus}^{\mathfrak{B}} \\ s \in \|\varphi\|_{\odot}^{\mathfrak{A}} &\text{ if and only if } s \in \|\varphi\|_{\odot}^{\mathfrak{B}} \end{aligned}$$

In other words

$$\|\varphi\|_{\#}^{\mathfrak{A}} = \|\varphi\|_{\#}^{\mathfrak{B}} \cap S(\mathfrak{A}).$$

for $\# \in \{\oplus, \ominus, \odot\}$.

Proposition 6.8 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QLFI1**-structures over Σ such that $\mathfrak{A} \subseteq \mathfrak{B}$. The following are equivalent:

- (a) $\mathfrak{A} \prec \mathfrak{B}$;
- (b) for every formula φ and every $s \in S(\mathfrak{A})$, we have

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi[s] \text{ if and only if } \mathfrak{B} \Vdash_{\mathbf{QLFI1}} \varphi[s].$$

Proof. Routine. ■

Definition 6.9 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QLFI1**-structures over Σ . An embedding $h : \mathfrak{A} \rightarrow \mathfrak{B}$ is called an elementary **QLFI1**-embedding of $\mathfrak{A}$ into $\mathfrak{B}$ if, for every formula φ and every $s \in S(\mathfrak{A})$,

$$\mathfrak{A} \models_{\mathbf{QLFI1}} \varphi[s] \text{ if and only if } \mathfrak{B} \models_{\mathbf{QLFI1}} \varphi[h \cdot s].$$

Proposition 6.10 Let $\mathfrak{A} \subseteq \mathfrak{B}$, and let $h : A \rightarrow B$ be given by $h(x) = x$. Then $\mathfrak{A} \prec \mathfrak{B}$ if and only if h is an **LFII**-elementary embedding.

Proof. Immediate from Definitions 6.7 and 6.9. ■

Proposition 6.11 $\mathfrak{A} \prec \mathfrak{B}$ implies $\mathfrak{A} \equiv \mathfrak{B}$.

Proposition 6.12 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **QLFI1**-structures over Σ with equality. If A is finite and $\mathfrak{A} \equiv \mathfrak{B}$, then $\mathfrak{A} \cong \mathfrak{B}$.

Proof. Suppose that $|A| = n$, i.e. $A = \{a_1, \dots, a_n\}$, and let σ_n be the sentence

$$(\exists v_1) \cdots (\exists v_n) \left(\left(\bigwedge_{1 \leq i, j \leq n} (\neg E(v_i, v_j) \wedge \circ E(v_i, v_j)) \right) \wedge (\forall v_{n+1}) \left(\bigvee_{1 \leq i \leq n} E(v_{n+1}, v_i) \right) \right).$$

Clearly $\mathfrak{A} \models \sigma_n$ and since $\mathfrak{A} \equiv \mathfrak{B}$, it follows that $\mathfrak{B} \models \sigma_n$, hence $|B| = n$, i.e. $B = \{b_1, \dots, b_n\}$.

We will now prove that

($\star$) there exists $b \in B$ such that $\langle \mathfrak{A}, a_1 \rangle \equiv \langle \mathfrak{B}, b \rangle$.

Assume that ($\star$) does not hold. Then, for every $b_i \in B$, $1 \leq i \leq n$, there exists $\varphi_i(a_1) \in \text{SENT}(\Sigma \cup \{a_1\})$ such that

$$\langle \mathfrak{A}, a_1 \rangle \models \varphi_i(a_1) \quad \text{AND} \quad \langle \mathfrak{B}, b_i \rangle \models \neg \varphi_i(a_1) \wedge \circ \varphi_i(a_1),$$

where $a_1^{\langle \mathfrak{B}, b_i \rangle} = b_i$. Then

$$\begin{aligned} & (\mathfrak{A}, a_1) \models_{\mathbf{QLFI1}} \bigwedge_{i \leq n} \varphi_i(a_1) \\ & \text{ONLY IF } \mathfrak{A} \models_{\mathbf{QLFI1}} \bigwedge_{i \leq n} \varphi_i(x)[s_x^{a_1}] & 6.3 \\ & \text{ONLY IF } \mathfrak{A} \models_{\mathbf{QLFI1}} (\exists x) \bigwedge_{i \leq n} \varphi_i(x) & 3.21(7) \\ & \text{ONLY IF } \mathfrak{B} \models_{\mathbf{QLFI1}} (\exists x) \bigwedge_{i \leq n} \varphi_i(x) & (b) \\ & \text{ONLY IF there exists } b_k \in B \text{ such that } \mathfrak{B} \models_{\mathbf{QLFI1}} \bigwedge_{i \leq n} \varphi_i(x)[s_x^{b_k}] & 3.21(7) \\ & \text{ONLY IF } (\mathfrak{B}, b_k) \models_{\mathbf{QLFI1}} \bigwedge_{i \leq n} \varphi_i(a_1) & 6.3 \end{aligned}$$

Hence there exists $k \leq n$ such that

$$(\mathfrak{B}, b_k) \models_{\mathbf{QLFI1}} \varphi_k(a_1) \quad \text{AND} \quad (\mathfrak{B}, b_k) \models_{\mathbf{QLFI1}} \neg \varphi_k(a_1) \wedge \circ \varphi_k(a_1)$$

which, in view of 4.1, is a contradiction, and consequently $(\star)$ holds. By repeated application of $(\star)$ we obtain

$$\langle \mathfrak{A}, a_1, \dots, a_n \rangle \equiv \langle \mathfrak{B}, b_1, \dots, b_n \rangle$$

where $\langle \mathfrak{A}, a_1, \dots, a_n \rangle$ and $\langle \mathfrak{B}, b_1, \dots, b_n \rangle$ are Σ_A -structures. Let $h : A \rightarrow B$ be defined by $h(a_i) = b_i$, then $\langle \mathfrak{B}, h(a) \rangle_{a \in A}$ is a model of $\text{Diag}(\mathfrak{A})$. By 6.6, h is an isomorphism. Therefore, $\mathfrak{A} \cong \mathfrak{B}$. ■

Definition 6.13 Let $\mathfrak{A}$ be an **LFII**-structure. We call the *elementary diagram* of $\mathfrak{A}$ the theory

$$\text{Th}(\mathfrak{A}, A) := \{ \varphi \in \text{SENT}(\Sigma_A) \mid (\mathfrak{A}, A) \Vdash_{\mathbf{QLFII}} \varphi \}$$

Proposition 6.14 Let $\mathfrak{A}$ and $\mathfrak{B}$ be two **LFII**-structures with equality over Σ , and let $h : A \rightarrow B$. The following are equivalent:

- (a) h is an elementary embedding of $\mathfrak{A}$ into $\mathfrak{B}$.
- (b) $(\mathfrak{B}, h(A))$ is an **LFII**-model of $\text{Th}(\mathfrak{A}, A)$.
- (c) There exists an extension $\mathfrak{C} \succeq \mathfrak{A}$ and an isomorphism $g : \mathfrak{A} \rightarrow \mathfrak{B}$ such that $g \supset h$.

Proof. “(a) only if (c)”. The **LFII**-structure $\mathfrak{C}$ and the isomorphism g are obtained as in 6.6. From 6.10 it follows that $\mathfrak{A} \preceq \mathfrak{C}$.

“(c) only if (a)”. Trivial.

“(a) only if (b)”. Let $\varphi \in \text{Th}(\mathfrak{A}, A)$ and consider $\psi(x_1, \dots, x_n) \in \text{For}(\Sigma)$ such that $\psi(a_1/x_1, \dots, a_n/x_n)$ is φ . Then

$$\begin{aligned} \langle \mathfrak{A}, A \rangle \Vdash_{\mathbf{QLFII}} \varphi & \text{ ONLY IF } \mathfrak{A} \Vdash_{\mathbf{QLFII}} \psi(x_1, \dots, x_n)[s], \text{ where } s(x_i) = a_i \\ & \text{ ONLY IF } \mathfrak{B} \Vdash_{\mathbf{QLFII}} \psi(x_1, \dots, x_n)[h \cdot s], \\ & \text{ ONLY IF } \langle \mathfrak{B}, h(A) \rangle \Vdash_{\mathbf{QLFII}} \varphi \end{aligned}$$

“(b) only if (a)”. Let $s \in S(\mathfrak{A})$, then

$$\begin{aligned} \mathfrak{A} \Vdash_{\mathbf{QLFII}} \varphi(x_1, \dots, x_n)[s] \\ \text{IFF } \langle \mathfrak{A}, A \rangle \Vdash_{\mathbf{QLFII}} \varphi[x_1/s(x_1), \dots, x_n/s(x_n)] \\ \text{IFF } \langle \mathfrak{B}, h(A) \rangle \Vdash_{\mathbf{QLFII}} \varphi[x_1/s(x_1), \dots, x_n/s(x_n)] \\ \text{IFF } \mathfrak{B} \Vdash_{\mathbf{QLFII}} \varphi(x_1, \dots, x_n)[h \cdot s] \end{aligned}$$

Lemma 6.15 Let $\Sigma = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$, $c \notin \mathcal{C}$, and $\Sigma' = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \cup \{c\} \rangle$. Let $\varphi(x)$ be a formula in Σ and let T be a theory in Σ . Then

$$T \Vdash_{\mathbf{QLFII}} \varphi(c) \text{ IFF } T \Vdash_{\mathbf{QLFII}} \forall x \varphi(x).$$

Proof. First, we must note that, since

$$T \subset SENT(\Sigma) \subset SENT(\Sigma'),$$

for every Σ -structure $\mathfrak{A}$ we have

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} T \text{ if and only if } (\mathfrak{A}, a) \Vdash_{\mathbf{QLFI1}} T \text{ for every } a \in A,$$

where $(\mathfrak{A}, a)$ is the extension such that $c^{(\mathfrak{A}, a)} = a$, and for every Σ' -structure $\mathfrak{A}$ we have

$$\mathfrak{A} \Vdash_{\mathbf{QLFI1}} T \text{ only if } \mathfrak{A}|_{\Sigma} \Vdash_{\mathbf{QLFI1}} T.$$

"ONLY IF". Suppose that $T \Vdash_{\mathbf{QLFI1}} \varphi(c)$. Let $\mathfrak{A}$ be a Σ -structure.

$$\begin{array}{llll} \mathfrak{A} \Vdash_{\mathbf{QLFI1}} T & \text{ONLY IF} & \text{for every } a \in A, (\mathfrak{A}, a) \Vdash_{\mathbf{QLFI1}} T & \\ & \text{ONLY IF} & \text{for every } a \in A, (\mathfrak{A}, a) \Vdash_{\mathbf{QLFI1}} \varphi(c) & \text{Hyp} \\ & \text{ONLY IF} & \text{for every } a \in A, \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi(x)[a] & \text{Thm} \\ & \text{ONLY IF} & \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \forall x \varphi(x) & \text{Def. of } \forall \end{array}$$

Hence

$$T \Vdash_{\mathbf{QLFI1}} \forall x \varphi(x).$$

"IF". Suppose that $T \Vdash_{\mathbf{QLFI1}} \forall x \varphi(x)$. Let $\mathfrak{A}$ be a Σ' -structure, and denote by $\mathfrak{A}|_{\Sigma}$ the reduct of $\mathfrak{A}$ to the signature Σ . Then

$$\begin{array}{llll} \mathfrak{A} \Vdash_{\mathbf{QLFI1}} T & \text{ONLY IF} & \mathfrak{A}|_{\Sigma} \Vdash_{\mathbf{QLFI1}} T & \\ & \text{ONLY IF} & \mathfrak{A}|_{\Sigma} \Vdash_{\mathbf{QLFI1}} \forall x \varphi(x) & \text{Hyp} \\ & \text{ONLY IF} & \text{for every } a \in A, \mathfrak{A}|_{\Sigma} \Vdash_{\mathbf{QLFI1}} \varphi(x)[a] & \text{Def. of } \forall \\ & \text{ONLY IF} & \text{for every } a \in A, (\mathfrak{A}|_{\Sigma}, a) \Vdash_{\mathbf{QLFI1}} \varphi(c) & \text{Thm} \\ & \text{ONLY IF} & \mathfrak{A} \Vdash_{\mathbf{QLFI1}} \varphi(c) & \end{array}$$

Hence, $T \Vdash_{\mathbf{QLFI1}} \varphi(c)$. ■

Lemma 6.16 Let $\Sigma = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \rangle$, $\{c_1, \dots, c_n\} \cap \mathcal{C} = \emptyset$, and $\Sigma' = \langle \mathcal{P}, \mathcal{F}, \mathcal{C} \cup \{c_1, \dots, c_n\} \rangle$. Let $\varphi(x_1, \dots, x_n)$ be a formula in Σ and let T be a theory in Σ . Then

$$T \Vdash_{\mathbf{QLFI1}} \varphi(c_1, \dots, c_n) \text{ IFF } T \Vdash_{\mathbf{QLFI1}} \forall x_1, \dots, \forall x_n \varphi(x_1, \dots, x_n).$$

Notation. Given signatures Σ , Σ_1 , and Σ_2 such that $\Sigma \subset \Sigma_1$ and $\Sigma \subset \Sigma_2$, if $\mathfrak{A}$ and $\mathfrak{B}$ are **LFII**-structures over Σ_1 and Σ_2 respectively, we write

$$\mathfrak{A} \equiv_{\Sigma} \mathfrak{B}$$

to indicate that the Σ -reducts of $\mathfrak{A}$ and $\mathfrak{B}$ are elementarily equivalent, and with the notation

$$f : \mathfrak{A} \prec_{\Sigma} \mathfrak{B}$$

we indicate that f is an elementary embedding of $\mathfrak{A}|_{\Sigma}$ into $\mathfrak{B}|_{\Sigma}$.

Theorem 6.17 Let $\mathfrak{A}$ be an **LFII**-structure over Σ , $\mathfrak{B}$ an **LFII**-structure over Σ' , and $\Sigma \subset \Sigma'$. If $\mathfrak{A} \equiv_{\Sigma} \mathfrak{B}$, then there exists an **LFII**-structure $\mathfrak{C}$ over Σ' such that

$$\begin{array}{ccc} \mathfrak{A} & & \\ & \searrow f & \\ \mathfrak{B} & \xrightarrow{g} & \mathfrak{C} \end{array}$$

where $f : \mathfrak{A} \prec_{\Sigma} \mathfrak{C}$ and $g : \mathfrak{B} \prec_{\Sigma'} \mathfrak{C}$.

Proof. Let $\Sigma'_{\mathfrak{A}\mathfrak{B}} := \Sigma_{\mathfrak{A}} \cup \Sigma_{\mathfrak{B}}$ (assuming $A \cap B = \emptyset$). Consider the theory

$$T := Th(\mathfrak{A}, A) \cup Th(\mathfrak{B}, B) \subset Sent(\Sigma'_{\mathfrak{A}\mathfrak{B}}).$$

Suppose that some finite $T_0 \subset T$ has no models. By taking conjunctions there exist $\varphi(\bar{a}) \in Th(\mathfrak{A}, A) \cap T_0$ and $\psi(\bar{b}) \in Th(\mathfrak{B}, B) \cap T_0$ such that $\varphi \wedge \psi$ has no models. It follows that no **LFII**-structure over $\Sigma'_{\mathfrak{A}\mathfrak{B}}$ satisfying $\psi(\bar{b})$ can be expanded to an **LFII**-structure with interpretations for the symbols $\bar{a}$ such that the expansion satisfies $\varphi(\bar{a})$. Consequently

$$\begin{aligned} Th(\mathfrak{B}, B) &\models \neg\varphi(\bar{a}) \wedge \circ\varphi(\bar{a}) \\ \Rightarrow Th(\mathfrak{B}, B) &\models \forall \bar{x}(\neg\varphi(\bar{x}) \wedge \circ\varphi(\bar{x})) && \text{Lemma on constants} \\ \Rightarrow (\mathfrak{B}, B) &\models \forall \bar{x}(\neg\varphi(\bar{x}) \wedge \circ\varphi(\bar{x})) \\ \Rightarrow (\mathfrak{A}, A) &\models \forall \bar{x}(\neg\varphi(\bar{x}) \wedge \circ\varphi(\bar{x})) \\ \Rightarrow (\mathfrak{A}, A) &\models \neg\varphi(\bar{a}) \wedge \circ\varphi(\bar{a}) \end{aligned}$$

Therefore, $(\mathfrak{A}, A) \models \varphi(\bar{a}) \wedge \neg\varphi(\bar{a}) \wedge \circ\varphi(\bar{a})$. It follows from the Compactness Theorem that T has a model $\mathfrak{C}$.

We define the maps $f : A \rightarrow C$ as $f(x) = x^{\mathfrak{C}}$, and $g : B \rightarrow C$ as $g(x) = x^{\mathfrak{C}}$. It follows from 6.14 that f and g are elementary embeddings. $\blacksquare$

Theorem 6.18 (Robinson's Consistency Theorem) Let Σ_1 and Σ_2 be two signatures, and let $\Sigma = \Sigma_1 \cap \Sigma_2$. If

- $T \subset Sent(\Sigma)$ is a complete theory,
- $T_1 \subset Sent(\Sigma_1)$ and $T_2 \subset Sent(\Sigma_2)$ extend T ,
- each of T_1 and T_2 has a model,

then $T_1 \cup T_2$ has a model.

Proof. Let

$\mathfrak{A}_0$ be a model of T_1 , $\mathfrak{B}_0$ be a model of T_2 . Then $\mathfrak{A}_0$ and $\mathfrak{B}_0$ are models of T , and since T is complete,

$$\mathfrak{A}_0 \equiv_{\Sigma} \mathfrak{B}_0$$

By virtue of 6.17 and 6.14, there exists a diagram

$$\begin{array}{ccc}
\mathfrak{A}_0 & & \\
& \searrow f_0 & \\
\mathfrak{B}_0 & \xrightarrow{\cong} & \mathfrak{B}_1
\end{array}$$

with $f_0 : \mathfrak{A}_0 \prec_{\Sigma} \mathfrak{B}_1$. Since $(\mathfrak{A}_0, A_0) \equiv_{\Sigma_{A_0}} (\mathfrak{B}_1, f_0(A_0))$, applying again 6.17 and 6.14, we obtain

$$\begin{array}{ccccc}
\mathfrak{A}_0 & \xrightarrow{\cong} & \mathfrak{A}_1 & & \\
& \searrow f_0 & & \uparrow g_0 & \\
\mathfrak{B}_0 & \xrightarrow{\cong} & \mathfrak{B}_1 & &
\end{array}$$

with $g_0 : \mathfrak{B}_1 \prec_{\Sigma} \mathfrak{A}_1$. Repeating this construction we obtain a tower

$$\begin{array}{ccccccc}
\mathfrak{A}_0 & \xrightarrow{\cong} & \mathfrak{A}_1 & \xrightarrow{\cong} & \mathfrak{A}_2 & \longrightarrow & \dots \\
& \searrow f_0 & \uparrow g_0 & \searrow f_1 & \uparrow g_1 & & \\
\mathfrak{B}_0 & \xrightarrow{\cong} & \mathfrak{B}_1 & \xrightarrow{\cong} & \mathfrak{B}_2 & \longrightarrow & \dots
\end{array}$$

with $f_k : \mathfrak{A}_k \prec_{\Sigma} \mathfrak{B}_{k+1}$ and $g_k : \mathfrak{B}_{k+1} \prec_{\Sigma} \mathfrak{A}_{k+1}$ such that

$$f_k \subset g_k^{-1} \subset f_{k+1} \quad ^4$$

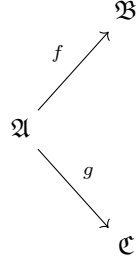
By the theorem on elementary chains, $\mathfrak{A} = \bigcup \mathfrak{A}_k$ is an **LFII**-model of T_1 and $\mathfrak{B} = \bigcup \mathfrak{B}_k$ is an **LFII**-model of T_2 . Moreover, $\bigcup f_k$ is an isomorphism of $\mathfrak{A}|\Sigma$ onto $\mathfrak{B}|\Sigma$. Thus, $\mathfrak{B}$ is isomorphic to an **LFII**-structure $\mathfrak{B}'$ such that $\mathfrak{A}|\Sigma = \mathfrak{B}'|\Sigma$. Let $\mathfrak{C}$ be the **LFII**-structure over $\Sigma_1 \cup \Sigma_2$ where

- $|\mathfrak{C}| = |\mathfrak{A}|$,
- If c is a constant symbol in Σ_2 , then $c^{\mathfrak{C}} := c^{\mathfrak{B}'}$.
If c is a constant symbol in $\Sigma_1 - \Sigma_2$, then $c^{\mathfrak{C}} := c^{\mathfrak{A}}$.
- If F is a function symbol in Σ_2 , $F^{\mathfrak{C}} = F^{\mathfrak{B}}$.
If F is a function symbol in $\Sigma_1 - \Sigma_2$, $F^{\mathfrak{C}} = F^{\mathfrak{A}}$.
- If P is a predicate symbol in Σ_2 , $P^{\mathfrak{C}} = P^{\mathfrak{B}'} = \langle P_{\oplus}^{\mathfrak{B}'}, P_{\ominus}^{\mathfrak{B}'}, P_{\odot}^{\mathfrak{B}'} \rangle$.
If P is a predicate symbol in $\Sigma_1 - \Sigma_2$, $P^{\mathfrak{C}} := P^{\mathfrak{A}} = \langle P_{\oplus}^{\mathfrak{A}}, P_{\ominus}^{\mathfrak{A}}, P_{\odot}^{\mathfrak{A}} \rangle$.

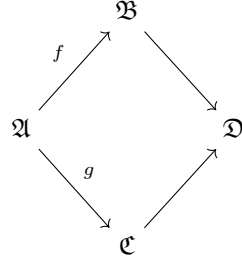
Then $\mathfrak{C}$ is a model of $T_1 \cup T_2$. ■

Theorem 6.19 (Amalgamation) Every diagram

⁴This follows from the characterization of embeddings by elementary diagrams.



of elementary embeddings between **LF11**-structures over Σ , can be completed to a commutative diagram



of elementary embeddings between **LF11**-structures over Σ .

Proof. We reformulate the elementary diagram $Th(\mathfrak{B}, B)$ in the signature

$$\Sigma_1 = \Sigma(A \cup (B - f(A))).$$

In this signature we have

$$c_a^{\mathfrak{B}} = f(a),$$

if $a \in A$. Similarly, we write the elementary diagram $Th(\mathfrak{C}, C)$ in the signature

$$\Sigma_2 = \Sigma(A \cup (C - g(A))).$$

In this signature we have

$$c_a^{\mathfrak{C}} = g(a),$$

if $a \in A$. Then $\Sigma_1 \cap \Sigma_2 = \Sigma(A)$. In Σ_A we have the complete theory $Th(\mathfrak{A}, A)$, and both $Th(\mathfrak{B}, B)$ and $Th(\mathfrak{C}, C)$ are extensions of it in Σ_1 and Σ_2 , respectively. By 6.18, $Th(\mathfrak{B}, B) \cup Th(\mathfrak{C}, C)$ has a model $\mathfrak{D}$.

We now define the functions

$$\begin{array}{ccc}
f^* : \mathfrak{B} & \rightarrow & \mathfrak{D} \\
x & \mapsto & c_x^{\mathfrak{D}}
\end{array}
\quad \text{and} \quad
\begin{array}{ccc}
g^* : \mathfrak{C} & \rightarrow & \mathfrak{D} \\
x & \mapsto & c_x^{\mathfrak{D}}
\end{array}$$

Since $(\mathfrak{D}, f^*(B)) \models_{\mathbf{QLF11}} Th(\mathfrak{B}, B)$ and $(\mathfrak{D}, g^*(C)) \models_{\mathbf{QLF11}} Th(\mathfrak{C}, C)$, it follows from 6.14 that f^* and g^* are elementary embeddings.

Let $a \in \mathfrak{A}$, then $f^*(f(a)) = f^*(c_a^{\mathfrak{B}}) = c_a^{\mathfrak{D}} = g^*(c_a^{\mathfrak{C}}) = g^*(g(a))$. Thus $f^*f = g^*g$, i.e. the diagram commutes. ■

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